

TOPICAL REVIEW

Leveraging AI for Intrusion Detection in IoT Ecosystems: A Comprehensive Study

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ABSTRACT The widespread adoption of Internet of Things (IoT) devices has brought about unprecedented connectivity and convenience, but it has also ushered in new security challenges. As IoT systems become integral parts of various domains, protecting them from malicious intrusions is crucial. This paper thoroughly examines how Artificial Intelligence (AI) techniques are applied to develop practical Intrusion Detection Systems (IDS) specifically designed for IoT environments. The examination comprehensively analyzes current state-of-the-art AI methodologies utilized in IoT-based IDS, including machine learning, deep learning, and anomaly detection algorithms. The paper delves into IoT systems' distinctive characteristics and vulnerabilities that demand specialized intrusion detection mechanisms. It assesses various AI-based approaches' performance, accuracy, and scalability in identifying and mitigating diverse cyber threats targeting IoT ecosystems. Beyond reviewing existing literature, the paper delves into the challenges of implementing AI-driven IDS in IoT, such as resource constraints, device heterogeneity, and dynamic network topologies. It also emphasizes potential research directions and solutions to tackle these challenges and strengthen the resilience of IoT systems against cyber threats. This all-encompassing survey aims to be a valuable resource for researchers, practitioners, and policymakers seeking to comprehend the current landscape of AI-based intrusion detection methods in IoT security. This paper contributes to the continuous efforts to fortify IoT ecosystems against evolving cybersecurity threats by shedding light on these approaches' strengths, limitations, and prospects.

INDEX TERMS Artificial intelligence (AI), intrusion detection systems (IDS), IoT systems, machine learning, deep learning, anomaly detection, cybersecurity.

I. INTRODUCTION

In recent years, the ubiquity of Internet of Things (IoT) systems has become a defining feature of the technological landscape. These interconnected networks, from smart thermostats to industrial sensors, promise increased efficiency, automation, and convenience in daily life. However, this widespread adoption of IoT brings inherent security challenges that require careful consideration. The pervasive integration of IoT into different facets of modern life has ushered in a new era of connectivity, convenience, and efficiency [1]. While these interconnected devices

enable seamless data exchange and unprecedented levels of automation, they also pose significant security challenges that threaten the integrity and functionality of IoT ecosystems [1].

The International Telecommunication Union officially introduced the concept of the IoT at the World Summit on the Information Society (WSIS) in 2005. IoT describes a distributed network that integrates diverse sensor devices and systems, such as sensor networks, RFID devices, barcode and QR code devices, global positioning systems, and others, with the Internet using both wired and wireless communication technologies. This integration allows embedded systems to communicate and interconnect.

As the use of IoT devices continues to rise, the vulnerabilities in these systems become increasingly apparent. The

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distributed nature of IoT, combined with device heterogeneity and the handling of vast amounts of sensitive data, creates a complex web of potential entry points for malicious actors. Cybersecurity concerns such as unauthorized access, data breaches, and denial-of-service attacks necessitate robust protective measures to secure the interconnected fabric of IoT [1].

A crucial element in defending against these emerging threats is the implementation of practical Intrusion Detection Systems (IDS). Unlike traditional security measures, IDS operates on the proactive detection of threats, aiming to identify and respond to potential security incidents in real-time. In the expansive and dynamic attack surface of IoT environments, the role of IDS becomes paramount in fortifying overall security [2].

This paper explores AI-based methods for implementing IDS in IoT systems, recognizing the transformative potential of Artificial Intelligence (AI) in enhancing security measures [3]. By leveraging advanced AI techniques such as machine learning, deep learning, and anomaly detection algorithms, the goal is to investigate how these methodologies can be tailored to the unique challenges presented by IoT [4], [5]. The overarching objective is to critically assess the current state-of-the-art AI-driven IDS for IoT, shedding light on their strengths, limitations, and potential avenues for improvement.

Intrusion Detection Systems (IDS) are pivotal in safeguarding modern networks by identifying and mitigating potential threats. As highlighted in [6], IDS are critical for maintaining system performance and ensuring financial stability. Effective intrusion detection helps minimize downtime and protect sensitive data, directly reducing costs associated with breaches and system recovery. Moreover, the study emphasizes that advancements in machine learning and artificial intelligence have significantly enhanced the accuracy and efficiency of IDS, making them indispensable in contemporary cybersecurity strategies.

A. IoT ARCHITECTURE

The architecture of IoT follows an entity-based structure divided into three layers as in figure 1.

- 1) Terminal Perception Layer: This layer collects data from various objects, including physical entities, IoT devices like sensors, RFID, and GPS devices, and access platforms connecting local IoT device networks to wide area networks.
- 2) Network Transport Layer: This layer transmits information from the terminal perception layer to the application service layer using non-cellular networks (such as ZigBee, Bluetooth, and Wi-Fi) and cellular networks (like NB-IoT and eMTC), handling data encoding, authentication, and transmission.
- 3) Application Service Layer: This layer processes data received from the network transport layer, integrating it with various industries to support vertical IoT

applications, such as smart grids, smart homes, and smart cities. It includes data storage, analysis, service provisioning, and operation maintenance and management subsystems.

B. APPLICATION IN AI-BASED IDS FOR IoT SYSTEMS

- 1) Data Collection and Preprocessing: Use statistical data cleaning and normalization methods. Employ time series models like ARIMA to detect temporal anomalies.
- 2) Feature Extraction and Selection: Apply SVD for dimensionality reduction and mutual information for selecting the most informative features.
- 3) Model Training: Utilize gradient descent for training machine learning models. Employ SVM for binary classification of normal versus abnormal behaviours.
- 4) Anomaly Detection: Implement unsupervised learning methods like clustering (e.g., k-means) to identify anomalies. Use entropy-based measures to detect irregularities in data distributions.
- 5) Real-Time Monitoring: Apply graph theory algorithms for network analysis and shortest path detection. Use real-time data processing techniques to monitor IoT network traffic for suspicious activities.
- 6) Continuous Improvement: Use reinforcement learning algorithms to adapt IDS models based on new data and detected threats, continuously improving detection accuracy and response times.

Understanding and applying these advanced mathematical concepts and formulas are crucial for developing robust AI-based IDS in IoT systems, enhancing their ability to detect and respond to security threats effectively.

To achieve this objective, the paper is organized as follows: the background section introduces the foundational concepts of AI, IDS, and IoT, providing a solid basis for further exploration. The literature review examines existing research, categorizing and analyzing various AI techniques applied to IoT-based IDS. The methodology section details the criteria and methods used in the literature review. Subsequent sections discuss specific applications of AI-based IDS in IoT, the challenges in their implementation, and potential future directions. The paper concludes with a summary of key findings, emphasizing the significance of AI in shaping the future of IoT security.

The paper is structured as follows: Section II provides a review of related work; Section III presents a classification and comparison of AI-based intrusion detection systems; Section V explains the Preliminary Mathematics for AI-based methods for implementing IDS in IoT; Section VI discusses the challenges faced by these systems; and Chapter VII concludes the paper.

II. LITERATURE REVIEW

Artificial Intelligence (AI) is a broad field in computer science where machines simulate intelligent behaviour,

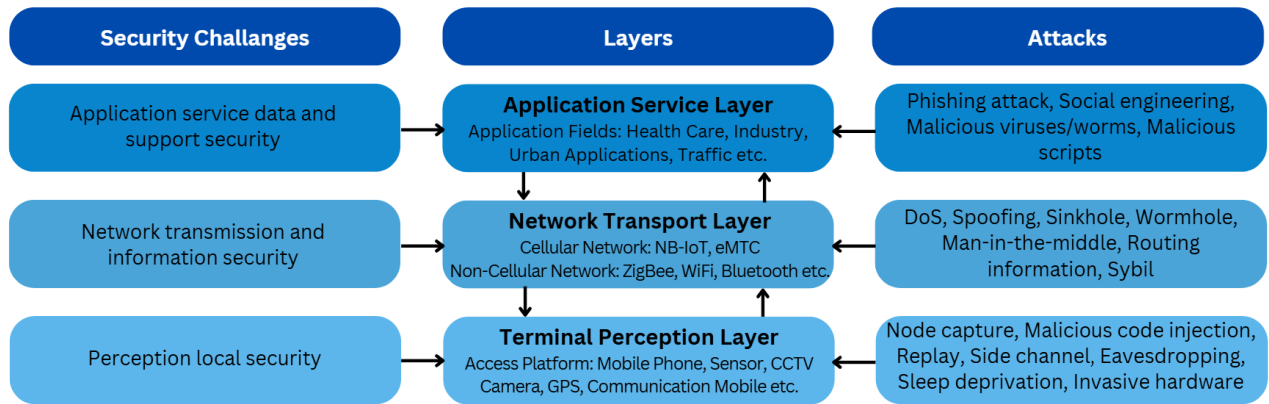


FIGURE 1. A comprehensive representation of security challenges, layers, and associated attacks in IoT systems.

employing various techniques to enable systems to perform tasks typically requiring human intelligence [2]. These tasks encompass learning from data (machine learning), decision-making based on rules and logic (expert systems), and understanding natural language (natural language processing). In cybersecurity, AI is crucial in developing advanced algorithms that enhance security measures, particularly in Intrusion Detection Systems (IDS) [2].

Intrusion Detection Systems (IDS) are security mechanisms designed to monitor and analyze network or system activities for signs of malicious behaviour or policy violations. Two primary types of IDS include Network-based IDS (NIDS), which scrutinizes network traffic for suspicious patterns, and Host-based IDS (HIDS), monitoring individual devices for anomalies or signs of compromise. IDS serves as a vital component of cybersecurity by providing early detection and response to potential threats, thereby aiding in the protection of critical systems and data [7].

The Internet of Things (IoT) refers to the interconnected network of physical devices, vehicles, appliances, and other items embedded with sensors, software, and connectivity features to collect, exchange, and act upon data. IoT systems are characterized by their vast heterogeneity, encompassing diverse devices with varying computational capabilities, communication protocols, and functionalities. They enable seamless communication and data exchange, facilitating automation and real-time decision-making across various domains, including healthcare, manufacturing, smart cities, and more [7].

However, the proliferation of IoT also introduces unique security challenges. The inherent characteristics of IoT, as in table 1, such as the massive scale of interconnected devices, diversity in device types and manufacturers, and the often resource-constrained nature of these devices, amplify the complexity of securing these systems. Additionally, IoT devices are frequently deployed in environments where security may not have been a primary consideration during design or where updates and patches are challenging to implement, making them attractive targets for cyber attackers [8].

Moreover, IoT devices' immense volumes of data, often containing sensitive information, pose significant privacy concerns. Data integrity, confidentiality, and availability become critical concerns in IoT environments, where a breach or manipulation of data can have severe consequences ranging from privacy violations to disruptions in critical services [8]. The detailed view of ML using IoT security is shown in figure 2.

Navigating these unique challenges and intricacies inherent in IoT environments requires a nuanced understanding of the interplay between AI-driven security mechanisms, IDS, and the diverse landscape of interconnected IoT devices. This understanding forms the foundation for exploring and developing effective strategies to safeguard IoT ecosystems against evolving cyber threats [9].

Addressing these unique characteristics and security challenges requires a holistic approach that combines robust authentication, encryption, standardized security practices, and adaptive security measures capable of accommodating the dynamic nature of IoT environments, as in table 2. As the IoT ecosystem evolves, staying ahead of emerging threats and vulnerabilities remains critical for ensuring these interconnected systems' security and resilience.

A. PRELIMINARY MATHEMATICS FOR AI-BASED METHODS FOR IMPLEMENTING IDS IN IoT

Implementing AI-based IDS in IoT systems requires advanced mathematical concepts and formulas across several domains. Here are some critical areas, along with their associated formulas [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22]:

1) PROBABILITY AND STATISTICS

- Bayes' Theorem

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} \quad (1)$$

TABLE 1. Unique characteristics of IoT environments.

Characteristics	Description
The heterogeneity of Devices	IoT encompasses various devices with different hardware, operating systems, and communication protocols. This heterogeneity poses challenges for implementing standardized security measures across the entire ecosystem [9].
Massive-scale	IoT networks often consist of interconnected devices, ranging from small sensors to large-scale industrial machinery. Managing security at such a scale requires scalable solutions and robust infrastructure.
Limited Resources	IoT devices have limited computational power, memory, and energy resources. Security solutions must be resource-efficient to operate effectively without imposing undue burdens on these devices [9].
Dynamic Network Typologies	IoT devices frequently operate in dynamic and ad-hoc network environments. Devices may join or leave the network; network typologies can change rapidly. Security mechanisms must adapt to these dynamic conditions [9].
Diverse Applications	IoT are deployed across various domains, including healthcare, transportation, smart homes, and industrial settings. Each application has unique security requirements and challenges, necessitating tailored approaches for effective protection.
Data Volume and Variety	IoT generate vast amounts of data, often of diverse types. Handling and securing this data, including sensitive information, requires robust data management and encryption practices.

TABLE 2. Security challenges in IoT environments.

Characteristics	Description
Inadequate Authentication and Authorization	some IoT devices may need more robust authentication mechanisms due to resource constraints. Weak or non-existent authentication can lead to unauthorized access, data breaches, and the compromise of entire IoT networks [10].
Insufficient Encryption	Ensuring the confidentiality of data transmitted between IoT devices is challenging. Inadequate encryption can expose sensitive information to eavesdropping and interception, leading to privacy violations.
Lack of Standardization	The absence of standardized security protocols across IoT devices and platforms creates vulnerabilities. A lack of uniform security standards hampers interoperability and makes it challenging to implement consistent security measures [10].
Physical Vulnerabilities	Many IoT devices are deployed in physically accessible environments, making them susceptible to tampering and physical attacks. This is particularly relevant in industrial settings and critical infrastructure.
Limited Update Mechanisms	IoT devices often lack efficient mechanisms for software updates and patches. This limitation can result in devices running outdated, vulnerable software, making them attractive targets for exploitation [10].
Privacy Concerns	The extensive collection of personal and sensitive data by IoT devices raises privacy concerns. Unauthorized access to this data can lead to identity theft, stalking, or other privacy breaches.
Denial-of-Service (DoS) Attacks	The massive scale of IoT networks makes them susceptible to DoS attacks, where attackers flood the network with traffic, rendering it inaccessible. Such attacks can disrupt critical services and compromise the functionality of IoT devices [10].

• Normal Distribution

$$f(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \quad (2)$$

• Maximum Likelihood Estimation

$$\hat{\theta} = \arg \max_{\theta} L(\theta|X) = \arg \max_{\theta} \prod_{i=1}^n f(x_i|\theta) \quad (3)$$

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2) LINEAR ALGEBRA

• Matrix Multiplication

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj} \quad (5)$$

• Eigenvalues and Eigenvectors

$$A\mathbf{v} = \lambda\mathbf{v} \quad (6)$$

where A is a matrix, $\mathbf{v}$ is an eigenvector, and λ is the corresponding eigenvalue.

• Singular Value Decomposition (SVD)

$$A = U\Sigma V^T \quad (7)$$

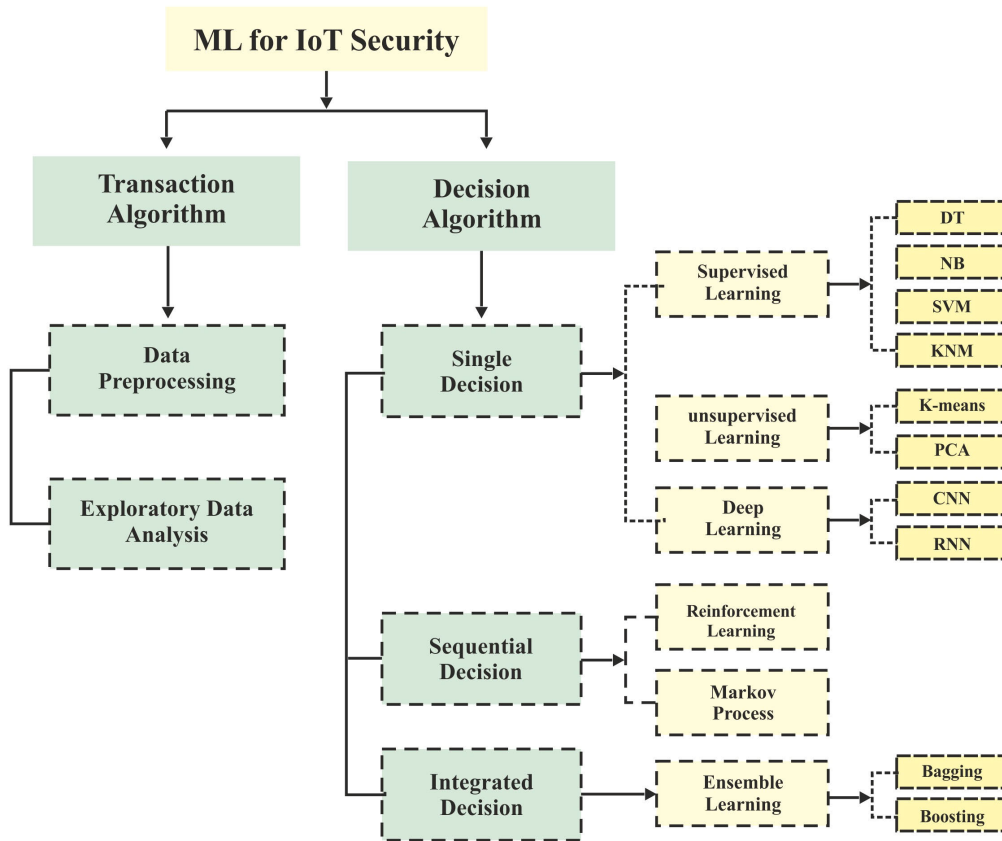


FIGURE 2. Machine learning for IoT security [11].

where A is a matrix, U and V are orthogonal matrices, and Σ is a diagonal matrix of singular values.

3) CALCULUS

- Gradient (Vector of Partial Derivatives)

$$\nabla f(\mathbf{x}) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right) \quad (8)$$

- Chain Rule

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x) \quad (9)$$

4) OPTIMIZATION

- Gradient Descent Update Rule

$$\theta := \theta - \eta \nabla J(\theta) \quad (10)$$

where θ represents the parameters, η is the learning rate, and $J(\theta)$ is the cost function.

- Lagrange Multipliers

$$\mathcal{L}(x, \lambda) = f(x) + \lambda(g(x) - c) \quad (11)$$

It's used to find a function's local maxima and minima subject to equality constraints.

5) GRAPH THEORY

- Adjacency Matrix

$$A_{ij} = \begin{cases} 1 & \text{if there is an edge from vertex } i \text{ to vertex } j \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

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$$A_{ij} = \begin{cases} 1 & \text{if there is an edge from vertex } i \text{ to vertex } j \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

- Dijkstra's Algorithm

$$\text{minimize } \sum_{(u,v) \in E} d(u,v) \quad (14)$$

The algorithm above is applicable for computing the shortest path within networked devices.

6) INFORMATION THEORY

- Entropy

$$H(X) = - \sum_i P(x_i) \log P(x_i) \quad (15)$$

- Mutual Information

$$I(X; Y) = \sum_{x \in X} \sum_{y \in Y} P(x, y) \log \frac{P(x, y)}{P(x)P(y)} \quad (16)$$

7) MACHINE LEARNING

- Linear Regression

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n \quad (17)$$

- Logistic Regression

$$P(y = 1|x) = \sigma(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n) \quad (18)$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function.

- Support Vector Machine (SVM)

$$\text{minimize } \frac{1}{2} \|\mathbf{w}\|^2 \text{ subject to } y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1 \quad (19)$$

8) TIME SERIES ANALYSIS

- Autoregressive Model (AR)

$$X_t = c + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + \epsilon_t \quad (20)$$

- Moving Average Model (MA)

$$X_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q} \quad (21)$$

- ARIMA Model

$$X_t = c + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t \quad (22)$$

B. MACHINE LEARNING IN INTRUSION DETECTION FOR IoTS

Machine Learning (ML) is a subset of artificial intelligence focused on creating algorithms and statistical models that enable computer systems to perform tasks without explicit programming. ML centres on empowering machines to learn patterns, make predictions, or make data-based decisions, evolving and improving performance through experience showing in the tables 3, 4, 5, 6, 7 and 8.

1) RELATED WORKS IN IoT INTRUSION DETECTION USING ML

Raoufi et al. [12] conducted a comprehensive review of deep learning (DL) applications in IoT environments, analyzing 56 articles published between 2019 and April 2024. Their study highlights the integration of DL techniques for processing complex and unstructured IoT data, identifying various tools and frameworks that facilitate this process. The authors emphasize the need for more scalable and robust DL models to enhance IoT data analysis.

Similarly, Rani and Niranjana [13] explored the role of machine learning in strengthening IoT security, categorizing ML/DL techniques based on their effectiveness in detecting and mitigating security threats. They also outline challenges and propose future research directions for improving IoT system security.

In another study, Prakash et al. [14] proposed a framework for anomaly detection in IoT systems using machine learning. Evaluating four ML algorithms—Logistic Regression, Linear Discriminant Analysis, Classification and Regression Trees, and Gaussian Naive Bayes—they demonstrated improved accuracy in identifying anomalies, thereby strengthening IoT security.

Li [15] focused on optimizing resource allocation in IoT networks through ML-driven techniques. His study analyzed strategies for both Low-Power and Mobile IoT Networks, showcasing how ML can enhance network efficiency and energy optimization.

Li et al. [16] examined the integration of cloud computing and ML in IoT monitoring and control systems. Their research demonstrates how ML, combined with cloud computing, can improve data processing efficiency and intelligent decision-making. They also address critical challenges such as latency and data privacy, proposing solutions to enhance IoT responsiveness and accuracy.

Musleh et al. [32] With the rise in IoT device usage, the paper emphasizes enhanced internet security, particularly in protecting vulnerable devices from malicious traffic. The focus is developing a Machine Learning (ML)-based Intrusion Detection System (IDS) for IoT, evaluating different feature extraction algorithms and ML models. The study assesses various feature extractors and ML algorithms, presenting a detailed evaluation using the IEEE Dataport dataset. Notably, VGG-16 combined with stacking achieves the highest accuracy of 98.3%.

Tekin et al. [17] The paper addresses security and privacy concerns in Smart Home Systems (SHSs) by proposing on-device ML-based Intrusion Detection Systems (IDS). It highlights the vulnerability of data shared with the cloud and introduces a comparative analysis of on-device ML algorithms regarding energy consumption for IoT intrusion detection applications. The Decision Tree (DT) algorithm on-device is identified as offering better results in training time, inference time, and power consumption.

Xu et al. [18] focusing on cybersecurity in the IoT, the research introduces a data-driven approach for intrusion and anomaly detection using ML. The study employs the Synthetic Minority Oversampling Technique (SMOTE) and mutual information to enhance the training dataset. Automated machine learning is utilized for optimal algorithm selection, achieving a multiclass classification accuracy of 99.7%, surpassing existing algorithms.

Shewale et al. [19] compare the performance of Multilayer Perceptron (MLP) and Long Short-Term Memory (LSTM) classifiers for Intrusion Detection Systems (IDS) in the context of the CICDDoS2019 dataset. Both classifiers exhibit high accuracy with minimal loss, with LSTM slightly outperforming MLP. The findings aim to assist security experts and researchers in selecting the most suitable ML algorithm for IDS based on specific requirements.

Savanović et al. [20] focus on the integration of the Internet of Things (IoT) into everyday life, particularly in healthcare. It addresses security challenges through ML algorithms optimized via a modified Firefly algorithm for detecting security issues in IoT devices used for Healthcare 4.0. The study emphasizes the contribution of metaheuristic solutions to sustainability in healthcare IoT systems.

Lin et al. [21], highlighting security challenges in IoT applications, the paper proposes the application of cloud computing and ML to improve the performance of an Intrusion Detection System (IDS) for IoT. The study introduces the Multi-Feature Extraction Extreme Learning Machine (MFE-ELM) algorithm for cloud computing, emphasizing low weights, short training time, and high detection accuracy. Simulation experiments demonstrate the effectiveness of the proposed algorithm in real-time intrusion detection for heterogeneous IoT data.

Sarhan et al. [33] addresses the unreliability of current Network Intrusion Detection Systems (NIDSs) in IoT networks, leading to security breaches. Researchers commonly explore untried combinations of Feature Reduction (FR) and Machine Learning (ML) techniques on diverse NIDS datasets. The study employs six ML models and evaluates three Feature Extraction (FE) algorithms on benchmark datasets, highlighting the need for a universal feature set.

Albulayhi et al. [23] The paper proposes a feature selection and extraction approach for anomaly-based Intrusion Detection Systems (IDSs) in IoT. Utilizing entropy-based methods, the study achieves high classification accuracy on IoT intrusion datasets using machine learning algorithms. Comparison with state-of-the-art studies demonstrates the superior performance of the proposed approach.

Saheed et al. [24] The paper discusses the security challenges posed by the proliferation of IoT devices, emphasizing the need for a machine learning-based intrusion detection system. The study applies ML-supervised algorithms on the UNSW-NB15 dataset, achieving competitive accuracy and highlighting the importance of addressing security and privacy concerns in IoT.

Faysal et al. [25] introduces a hybrid machine learning scheme, XGB-RF, for detecting intrusion attacks on IoT devices. Evaluating performance on the N-BaIoT dataset, the study demonstrates the model's effectiveness in detecting botnet attacks, outperforming existing algorithms in various metrics.

Rehman et al. [26] emphasizes the significance of data security in the IoT and proposes a machine learning-based intrusion detection scheme. The study reviews literature on IoT attacks and machine learning-based intrusion detection, highlighting the dynamic nature of attacks and the need for adaptive security measures.

Esmaili et al. [27] explores deep learning-based intrusion detection systems for securing IoT, specifically addressing DDoS attacks. The study recommends BiLSTM models for their effectiveness in classifying attacks in IoT and identifies

the need for further research on the impact of different data processing techniques.

Saif et al. [28] presents a hybrid intelligent Intrusion Detection System (HIIDS) for IoT-based healthcare applications. Using metaheuristic algorithms and supervised learning, the study achieves high accuracy in detecting security attacks on cloud servers, contributing to reducing security concerns in IoT systems.

Imad et al. [29] focuses on using machine learning techniques for Intrusion Detection Systems (IDS) in the IoT network. The study preprocesses data, utilizes random forest for feature extraction, and compares various classification techniques, with the Decision Tree model achieving the highest accuracy of 100%.

Salman et al. [30] proposes a framework for IoT device identification and malicious traffic detection using machine learning. The framework achieves high accuracy for device-type identification, traffic-type classification, and abnormal traffic detection, contributing to IoT network security.

Roy et al. [31] introduces a novel intrusion detection model using machine learning for IoT environments. The model achieves high detection rates and low false alarm rates through optimizations, outperforming existing models while being lightweight and suitable for resource-constrained IoT networks.

Sarhan et al. [33] focusing on IoT network security, the paper explores feature selection's importance for ML-based network intrusion detection systems. Three feature selection algorithms are compared, and the results show the potential for reducing computational and storage costs while maintaining detection accuracy, which is crucial for resource-limited IoT systems.

Verma et al. [34] addresses IoT security vulnerabilities and proposes a binary classifier approach using a gradient boosting machine (GBM) ensemble for malicious traffic filtering. The model achieves high positive class performance metrics, including 98.27% accuracy, making it effective against cyber threats in critical IoT applications. Rahman et al. [35] investigating the growth of IoT devices, the paper proposes a feature selection technique for intrusion detection using deep learning and machine learning. The approach achieves a high detection accuracy of up to 99.95%, showcasing its effectiveness on the Aegean Wi-Fi Intrusion Dataset.

Gad et al. [36] focuses on securing Vehicular Ad Hoc Networks (VANETs) using ML methods for intrusion detection. Using the ToN-IoT dataset, various ML methods are tested, with XGBoost outperforming others, addressing the need for security in VANETs.

Saba et al. [37] addressing IoT network security, the paper proposes a two-stage hybrid method using genetic algorithms and ML algorithms (SVM, ensemble classifier, decision tree). The achieved accuracy is 99.8%, emphasizing accuracy's importance in enhancing intrusion detection systems.

Dina and Manivannan [38] offer a comprehensive survey of ML-based intrusion detection approaches over the last

ten years, discussing the classification of legitimate and anomalous behavior using ML techniques and highlighting open issues in the field.

Manhas and Kotwal [39] emphasizes the importance of data security; the paper evaluates different ML techniques for implementing an Intrusion Detection System (IDS). Decision tree emerges as the best classifier, demonstrating the effectiveness of ML in network security.

Krishnan et al. [40] explore using supervised learning algorithms to detect malicious traffic in IoT systems. Various feature selection methods and classifiers are employed, achieving high accuracy in predicting malicious network traffic against IoT devices.

De Carvalho Bertoli et al. [41] introduce the AB-TRAP framework for network intrusion detection, considering updated network traffic and addressing operational concerns. The framework achieves a high f1-score and area under the ROC curve, offering a reproducible and updated approach to intrusion detection system deployment.

Verma and Ranga [61] focus on the increasing popularity of Internet of Things (IoT) research and its susceptibility to cyber threats, notably denial of service (DoS) attacks. They propose using machine learning classification algorithms for securing IoT against DoS, conducting a comprehensive study on classifiers, benchmarking with popular datasets, and evaluating response time on IoT-specific hardware.

Sai Kiran et al. [43] address the vulnerability of IoT networks to attacks, emphasizing the need for machine learning models to identify attacks. They create a test bed using Node MCU ESP8266, a DHT11 sensor, and a wireless router, conducting attacks like Man in the Middle using ARP Poisoning. Various machine learning classifiers are built to categorize data into standard and attack classes.

Rahman et al. [44] discuss the challenges of intrusion detection systems (IDS) for the expanding IoT in smart cities. They propose semi-distributed and distributed methods using machine learning models to overcome the limitations of centralized IDS. The study demonstrates promising results with comparable detection accuracy and highlights trade-offs between accuracy and building time performance.

Hindy et al. [45] focus on the MQTT protocol in IoT communication and the lack of IDS datasets for MQTT-based attacks. They evaluate six machine learning techniques using a simulated dataset, emphasizing the importance of flow-based features for discriminating attacks from benign traffic.

Amouri et al. [46] reiterate the significance of IoT in cybersecurity research, explicitly addressing MQTT-based attacks. They evaluate machine learning techniques using different abstraction levels of features and release a simulated MQTT dataset for further study.

Sadikin and Kumar [47] introduce an IDS for ZigBee-based IoT systems, combining rule-based and machine learning-based anomaly detection. They demonstrate methods to

detect known attacks and potential new types of attacks, emphasizing the hybrid approach's effectiveness.

Maleh [48] discuss the transformation of aerospace cyber-physical systems (CPS) into IoT, emphasizing the security threats. Machine learning techniques are applied to detect IoT attacks using a simulated dataset generated with the Cooja IoT simulator, showing promising accuracy, efficiency, and low energy consumption results.

Liu et al. [49] address the security vulnerabilities in IoT systems using machine learning algorithms to detect attacks in the NSL-KDD dataset. They find that tree-based and ensemble methods outperform other algorithms, with XGBoost achieving high accuracy, Matthews correlation coefficient, and area under the curve performance.

Bagaa et al. [50] present an ML-based security framework for IoT, leveraging Software Defined Networking (SDN) and Network Function Virtualization (NFV) for threat mitigation. The framework combines supervised learning and distributed data mining for efficient attack detection, demonstrating high accuracy in imaginative building scenarios.

Chaabouni et al. [66] addresses IoT security threats, mainly focusing on network intrusion detection systems (NIDS). It classifies IoT security threats, evaluates existing defense techniques, and reviews NIDS implementations and datasets. The survey extensively covers traditional and machine learning-based NIDS techniques, providing a comprehensive resource for researchers and industry professionals.

Papafotikas and Kakarountas [52] focus on protecting intelligent IoT devices against malware and attacks. It proposes a digital system with a Machine Learning-based clustering algorithm for detecting suspicious behavior in intelligent IoT devices. The K-Means Clustering Algorithm with Supervised Training is used to show successful detection of suspicious behavior.

Hasan et al. [53] address the rising concern of attacks and anomalies in IoT infrastructure. It compares the performance of machine learning models, including Logistic Regression, SVM, Decision Tree, Random Forest, and Artificial Neural Network, in predicting attacks and anomalies. Decision Tree, Random Forest, and ANN show promising results with 99.4% test accuracy.

Fenanir et al. [54] To enhance IoT security, the study proposes a lightweight intrusion detection system (IDS) using feature selection and classification. The research evaluates machine learning techniques, including Logistic Regression, Naive Bayes, Decision Tree, Random Forest, k-nearest Neighbor, SVM, and MLP, selecting the Decision Tree for its outstanding performance.

Sapre et al. [55] conduct a robust comparison of two widely used intrusion detection datasets, KDDCup99 and NSL-KDD, using various machine learning classifiers. The NSL-KDD dataset is of higher quality, achieving an average accuracy improvement of 20.18% over the KDDCup99 dataset.

Anthi et al. [56] proposes a three-layer intrusion detection system (IDS) for IoT networks using a supervised approach. It classifies IoT devices, identifies malicious packets during attacks, and classifies the type of attack. The proposed IDS achieves high F-measure results, demonstrating effective detection of IoT network attacks.

Jan et al. [57] focusing on denial-of-service (DoS) attacks in IoT networks, the paper develops a lightweight attack detection strategy using a supervised machine learning-based support vector machine (SVM). Simulation results show satisfactory performance in terms of classification accuracy and detection time.

Cassales et al. [58] introduce three data stream novelty detection techniques for intrusion detection in IoT networks. It proposes the IDSA-IoT architecture, combining edge network resources and a public cloud for efficient adaptation to emerging attack patterns.

Yao et al. [59] addressing the security of the Industrial Internet of Things (IIoT), the study surveys detection methods and system architectures of intrusion detection systems (IDS). It proposes a hybrid IDS architecture and a machine-learning-aided detection method, outperforming existing literature in various benchmarks.

Zolanvari et al. [60] emphasize securing IoT devices and exploring machine learning and big data analytics for IoT and IIoT security. It conducts a cyber-vulnerability assessment, reviews intrusion detection solutions using ML models, and presents a case study with a real-world testbed, demonstrating the effectiveness of ML-based anomaly detection against cyber-attacks.

Machine Learning (ML) plays a critical and nuanced role in bolstering Intrusion Detection Systems (IDS) within the specific context of Internet of Things (IoT) systems. The application of ML techniques addresses the challenges posed by IoT environments' vast and dynamic nature, where many interconnected devices generate diverse data types in real-time. This detailed exploration elucidates the multifaceted aspects of the role of ML in IoT intrusion detection. One of the fundamental contributions of ML in IoT-based IDS lies in its ability to recognize complex patterns and anomalies within the diverse and voluminous data generated by IoT devices. Traditional rule-based systems often struggle to adapt to the evolving attack landscape, making ML's pattern recognition capabilities particularly valuable. By learning from historical data, ML models discern the typical behaviours of interconnected devices, enabling the identification of abnormal patterns indicative of potential security threats [62].

Anomaly detection, a cornerstone of ML-based IDS, finds enhanced significance in the context of IoT systems. Establishing a baseline of normal behaviour allows ML algorithms to identify deviations that may signify potential intrusions. In IoT, where the attack surface is dynamic and continually changing, the adaptability of ML to evolving threats ensures that the IDS can effectively discern anomalies without relying

on predefined signatures. This adaptability reduces false positives, a crucial aspect in maintaining the efficacy of intrusion detection in the complex and heterogeneous IoT landscape. The role of ML in IoT intrusion detection extends beyond recognizing known threats. ML algorithms, particularly those utilizing unsupervised learning, can remarkably adapt to unknown or zero-day threats. In the rapidly evolving IoT ecosystem, where new vulnerabilities are discovered, unsupervised ML provides a proactive defence mechanism that can autonomously adapt to emerging threats without requiring manual intervention [63]. Behavioural analysis, facilitated by ML, is a critical component in the arsenal of an IDS for IoT systems. ML models can identify deviations that may signify potential security incidents by learning and understanding typical device behaviour patterns. This is particularly relevant in IoT environments where a diverse range of devices, each with unique behaviour, operates in tandem. ML's ability to discern abnormal behaviours contributes to proactively identifying potential security threats in this complex ecosystem [63].

Real-time threat detection is paramount in IoT systems where the speed of response is critical. ML-based IDS can operate in real-time, continuously analyzing the influx of data from IoT devices and making instantaneous decisions regarding the likelihood of a security threat. This capability enables swift incident response, reducing the potential impact of security breaches in the dynamic and interconnected IoT environment. Reducing false positives is a persistent challenge in intrusion detection, and ML provides an effective solution. ML algorithms, when well-trained, can significantly reduce false positives by accurately distinguishing between normal and malicious activities. This is crucial in IoT, where data's sheer volume and diversity can lead to a higher likelihood of false positives. ML's ability to discern nuanced patterns contributes to the precision of intrusion detection, allowing security teams to focus on genuine threats [64].

In dynamic IoT network environments, ML-based IDS showcases enhanced adaptation capabilities. Devices may join or leave the network, and network topologies can change rapidly. ML's adaptability ensures that the IDS remains effective in evolving IoT environments, where the landscape may constantly shift due to device additions, removals, or alterations in communication patterns [64]. In summary, the role of ML in Intrusion Detection within IoT systems is multifaceted, encompassing pattern recognition, anomaly detection, adaptability to unknown threats, behavioural analysis, real-time threat detection, and the reduction of false positives. ML is crucial in securing the intricate and ever-evolving IoT landscape, providing intelligent and adaptive defence mechanisms against various cyber threats [64].

2) MACHINE LEARNING ALGORITHMS IN IDS

In the realm of Machine Learning (ML) for Intrusion Detection Systems (IDS) in Internet of Things (IoT) systems,

various algorithms are employed to analyze network traffic, detect anomalies, and identify potential security threats. These algorithms leverage ML principles to learn from data, adapt to changing environments, and distinguish between normal and malicious activities. Below are detailed explanations of key ML algorithms used in the context of intrusion detection in IoT systems [65]:

1) Supervised Learning Algorithms

- a) Support Vector Machines (SVM): SVM is a powerful classification algorithm that finds the hyperplane that best separates data into different classes. In the context of IDS, SVM is employed to learn from labelled datasets where instances of normal and malicious network behaviour are explicitly identified. SVM is effective when the data exhibits clear separability. It creates a decision boundary that optimally distinguishes between normal and malicious activities. SVM's ability to handle high-dimensional data makes it suitable for analyzing diverse features in IoT network traffic [65].
- b) Decision Trees: Decision trees are hierarchical structures where each internal node represents a decision based on specific features, leading to subsequent nodes or leaves. These trees are constructed during the training phase to facilitate classification in intrusion detection. Decision trees in IDS can be interpreted as a series of 'if-else' rules, providing transparency in decision-making. In IoT, where interpretability is crucial, decision trees can be valuable for understanding the features contributing to identifying malicious activities [65].

2) Unsupervised Learning Algorithms:

- a) K-Means Clustering: K-Means is a clustering algorithm that groups data points into 'k' clusters based on similarity. Each cluster represents a group of data points that share similar characteristics. In IoT intrusion detection, K-Means can be applied to group network traffic data into clusters. Deviations from the patterns established by these clusters may indicate anomalous behaviour, potentially signalling an intrusion. K-Means is particularly useful when the expected behaviour of the network is not explicitly defined [66].
- b) Isolation Forests: Isolation Forests are anomaly detection algorithms that isolate instances considered anomalies. They achieve this by creating random decision trees and identifying cases that require fewer splits to be isolated. Isolation Forests are well-suited for detecting outliers in IoT network data. In intrusion detection, instances that deviate from the norm or exhibit unusual patterns are isolated, aiding in identifying potential security threats. The randomness in

tree construction makes it adaptable to IoT environments' dynamic and diverse nature [66].

These detailed explanations underscore the versatility of supervised and unsupervised learning algorithms in IoT intrusion detection. The choice of algorithm depends on factors such as the nature of the data, the availability of labelled samples, and the specific requirements of the IoT environment being monitored. Applying these ML algorithms contributes to creating robust and adaptive intrusion detection systems capable of securing IoT ecosystems against various threats.

3) FINDINGS AND DISCUSSION

The research on Intrusion Detection Systems (IDS) within Internet of Things (IoT) security indicates a rapidly changing landscape. Various studies have delved into and compared different machine learning (ML) techniques for identifying attacks and anomalies in IoT networks. These investigations encompassed lightweight IDS solutions, hybrid approaches combining rule-based and ML-based methods, and surveys assessing traditional and ML-based Network Intrusion Detection Systems (NIDS). There was a consistent emphasis on feature selection and extraction methods to enhance detection accuracy and reduce computational costs.

The exploration extended to diverse IoT applications, including healthcare, ZigBee-based systems, and Industrial IoT (IIoT), focusing on addressing unique security challenges. The proposed ML models consistently demonstrated high accuracy, often exceeding 99%, with Decision Trees, Random Forest, and ensemble learning approaches emerging as effective classifiers. Additionally, researchers investigated energy efficiency, privacy concerns, and reliability issues associated with NIDS in IoT networks. In summary, these findings underscore the importance of ML-based IDS in strengthening the security of IoT ecosystems, presenting promising results for efficient and accurate intrusion detection across various IoT domains.

Examining the findings brings crucial insights and implications for IoT security and Intrusion Detection Systems (IDS). The studies collectively underscore the increasing significance of ML techniques, spanning from decision trees to deep learning models, in bolstering the security of IoT ecosystems. The focus on feature selection and extraction methods highlights the importance of optimizing computational resources while upholding high detection accuracy. Addressing specific IoT applications, such as healthcare, ZigBee, and Industrial IoT, reflects a nuanced understanding of unique security challenges in each domain.

Considerations like energy consumption in on-device ML algorithms, privacy concerns in ML-supervised IDS, and the inadequacy of traditional NIDS in IoT networks contribute additional dimensions to the broader discourse on

TABLE 3. Year-wise literature survey on IoT using machine learning (2024).

Authors	Key Concept	Findings
Parisa Raoufi et. al [12]	Reviews DL applications in IoT, analyzing 56 articles from 2019–2024	Highlights DL's role in IoT data analysis, discusses tools, and identifies future research directions
Vakula R et. al [13]	Examines ML/DL techniques for IoT security threat detection	Categorizes ML approaches, evaluates effectiveness, and discusses challenges in IoT security implementation.
Vijay Prakash et. al [14]	Proposes an ML-based anomaly detection framework for IoT security	Evaluates ML models for anomaly detection, improving IoT network security.
Zhengdong Li [15]	Explores ML applications in optimizing IoT resource allocation	Analyzes ML's impact on network efficiency and energy optimization.
Hanzhe Li et. al [16]	Discusses ML and cloud computing integration for IoT monitoring and control	Demonstrates improved IoT responsiveness and accuracy while addressing latency and privacy challenges.

TABLE 4. Year-wise literature survey on IoT using machine learning (2023).

Authors	Key Concept	Findings
Musleh D. et al. [17]	ML-based IDS for IoT, Feature Extraction	Emphasis on IoT security, development of ML-based Intrusion Detection System (IDS) for IoT, evaluation of feature extraction algorithms, VGG-16 with stacking achieves highest accuracy of 98.3%.
Tekin N. et al. [18]	On-device ML for SHSs, Energy Consumption	Proposal of on-device ML-based IDS for Smart Home Systems (SHSs), comparative analysis of on-device ML algorithms in terms of energy consumption for IoT intrusion detection, Decision Tree algorithm yields optimal results.
Xu H. et al. [19]	Data-driven IDS for IoT, SMOTE, AutoML	Introduction of a data-driven approach for intrusion and anomaly detection in IoT, use of SMOTE and mutual information for dataset enhancement, Automated Machine Learning (AutoML) achieves multiclass classification accuracy of 99.7%.
Shewale Y. et al. [20]	MLP and LSTM for IDS, CIDDoS2019 dataset	Comparative study of MLP and LSTM classifiers for Intrusion Detection Systems (IDS), both classifiers exhibit high accuracy with minimal loss, LSTM slightly outperforms MLP.
Savanović N. et al. [21]	ML for Healthcare IoT, Metaheuristics	Application of ML algorithms optimized via a modified Firefly algorithm for detecting security issues in IoT devices used for Healthcare 4.0, emphasis on metaheuristic solutions contributing to sustainability in healthcare IoT systems.
Lin H. et al. [22]	Cloud-based ML for IoT IDS, MFE-ELM algorithm	Proposal of cloud-based ML for Intrusion Detection System (IDS) in IoT, Introduction of MFE-ELM algorithm for cloud computing, emphasis on low weights, short training time, and high detection accuracy.

IoT security. Moreover, the proposed lightweight models, hybrid approaches, and optimized frameworks contribute to the ongoing conversation on crafting efficient and resource-conscious IDS solutions for diverse IoT environments. Collectively, the findings reinforce the pivotal role of ML in elevating the security posture of IoT networks and underscore the necessity for adaptive and context-aware intrusion detection mechanisms.

C. DEEP LEARNING IN INTRUSION DETECTION FOR IoTS

Deep learning (DL) plays a crucial role in intrusion detection for IoT systems by effectively identifying and mitigating security threats. Traditional rule-based methods struggle with the complexity and volume of IoT data, making DL a powerful alternative. Techniques such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and autoencoders analyze network traffic patterns, detect anomalies, and classify potential threats in real time. DL-based intrusion detection systems (IDS) enhance IoT security by adapting to evolving cyber threats, reducing false alarms, and improving detection accuracy as been compared in tables 9, 10, 10, 11, 12, 13 and 14.

1) RELATED WORKS IN DEEP LEARNING

Raoufi et al. [12] conducted a comprehensive review of deep learning applications in IoT, analyzing 56 articles from 2019 to April 2024. They identified current challenges and potential breakthroughs at the intersection of deep learning and IoT, providing a roadmap for future research.

Vijayarajeswari et al. [67] developed an optimized deep learning framework for real-time intrusion and anomaly detection in IoT networks. Their system achieved high accuracy (98.5%) and low latency (50 ms) in detecting anomalies, with a 40% reduction in energy consumption, thus enhancing security and efficiency in large-scale IoT deployments.

Liu and Yang [68] presented a literature review and implementation of ensemble deep learning models for IoT network traffic anomaly detection. Their approach demonstrated over 98% accuracy in detecting anomalies using ensemble techniques, highlighting the effectiveness of deep learning in monitoring complex IoT networks.

Singh et al. [69] explored deep learning techniques for anomaly detection in IoT systems across various applications. They discussed the integration of models like CNNs and

TABLE 5. Year-wise literature survey on IoT using machine learning (2022).

Authors	Key Concept	Findings
Sarhan, M. et al. [23]	Unreliability of NIDS in IoT, ML for NIDS	Current NIDS unreliability in IoT networks, active research for improving NIDS technologies, exploration of untried FR and ML techniques, need for a universal feature set.
Albulayhi, K. et al. [24]	Feature selection for IoT IDS, ML algorithms	Proposal of a feature selection and extraction approach for anomaly-based IDS in IoT, use of entropy-based methods, high classification accuracy on IoT intrusion datasets, superiority over state-of-the-art studies.
Saheed, Y. K. et al. [25]	ML-supervised IDS for IoT, Privacy concerns	Introduction of ML-based IDS for detecting IoT network attacks, focus on privacy and security issues, application of ML algorithms on the UNSW-NB15 dataset, competitive results in various metrics, emphasis on addressing security and privacy concerns in IoT.
Faysal J. A. et al. [26]	XGB-RF for IoT intrusion detection, Botnet	Introduction of a hybrid ML scheme (XGB-RF) for detecting intrusion attacks on IoT devices, evaluation on the N-BaIoT dataset, superior performance in various metrics, effective detection of botnet attacks.
Rehman, E. et al. [27]	Machine learning for IoT security, Literature	Emphasis on data security in IoT, the Proposal of a machine learning-based intrusion detection scheme, review of literature on IoT attacks and ML-based intrusion detection, highlighting the dynamic nature of attacks, need for adaptive security measures.
Esmacili M. et al. [28]	Deep learning-based IDS, DDoS in IoT	Exploration of meta innovations using deep learning-based IDS, focus on BiLSTM models, the recommendation for further research on the impact of data processing techniques, identification of BiLSTM as a reliable solution for evaluating DDoS attacks in IoT.
Saif S. et al. [29]	Hybrid intelligent IDS for IoT healthcare	Introduction of a hybrid intelligent IDS for IoT-based healthcare applications, use of metaheuristic algorithms and supervised learning, high accuracy in detecting security attacks on cloud servers, contribution to reducing security concerns in IoT systems.
Imad M. et al. [30]	ML for IDS in IoT, Feature extraction	Use of machine learning techniques for IDS in IoT, preprocessing of data, random forest for feature extraction, comparison of classification techniques, Decision Tree model achieving the highest accuracy of 100%.
Salman O. et al. [31]	ML for IoT devices identification, Framework	Proposal of a framework for IoT devices identification and malicious traffic detection using ML, comparison of ML algorithms, achievement of high accuracy for device-type identification, traffic-type classification, and abnormal traffic detection.
Roy, S. et al. [32]	Lightweight ML IDS for IoT, Optimization	Introduction of a novel intrusion detection model using ML for IoT environments, optimization techniques including removal of multicollinearity, sampling, and dimensionality reduction, high detection rates, low false alarm rates, outperformance of existing models, lightweight and suitable for resource-constrained IoT networks.

TABLE 6. Year-wise literature survey on IoT using machine learning (2021).

Authors	Key Concept	Findings
Atul D. J. et al. [33]	Energy Aware Smart Home (EASH) framework	Addressing challenges in Cyber-Physical Systems (CPS), achieving 85% accuracy in communication failure and network attack analysis.
Sarhan M. et al. [34]	Feature Selection for IoT Network Security	Investigating feature selection's importance in ML-based intrusion detection systems, the potential for cost reduction while maintaining accuracy.
Verma P. et al. [35]	Binary Classifier for IoT Security	Proposing a binary classifier approach with GBM ensemble for filtering malicious traffic, achieving 98.27% accuracy in IoT network security.
Rahman M. A. et al. [36]	Feature Selection for Wi-Fi Intrusion	Utilizing feature selection with deep learning and ML for intrusion detection, achieving high accuracy of up to 99.95% on Aegean Wi-Fi Intrusion Dataset.
Gad, A. R. et al. [37]	ML Methods for VANET Security	Securing Vehicular Ad Hoc Networks (VANETs) using various ML methods, with XGBoost outperforming, addressing security needs in VANETs.
Saba, T. et al. [38]	Two-Stage Hybrid Method for IoT Security	Introducing a two-stage hybrid method using genetic algorithms and ML algorithms, achieving 99.8% accuracy in intrusion detection for IoT.
Dina A. S. et al. [39]	ML-Based Intrusion Detection Survey	Providing a comprehensive survey of ML-based intrusion detection approaches, discussing the classification of legitimate and abnormal behaviour.
Manhas J. et al. [40]	ML Techniques for IDS Implementation	Evaluating different ML techniques for implementing an Intrusion Detection System (IDS), highlighting decision tree as the best classifier.
Krishnan S. et al. [41]	Supervised Learning for IoT Malicious Traffic	Exploring the use of supervised learning algorithms for detecting malicious traffic in IoT systems, achieving high accuracy.
Bertoli G. D. C. et al. [42]	AB-TRAP Framework for Network Intrusion	Introducing the AB-TRAP framework for network intrusion detection, achieving high f1-score and ROC curve, offering an updated and reproducible approach.

LSTMs for real-time anomaly detection, emphasizing the need for adaptive learning techniques to enhance IoT security.

Ansar et al. [70] proposed an intrusion detection system for IoT environments using a hybrid CNN-LSTM deep learning model. Their system achieved 99.52% accuracy in classifying network traffic as benign or malicious, offering a robust solution for enhancing IoT network security.

Qaddoura et al. [80] attracted cybercriminal attention, leading to increased attacks. This paper introduces a Deep

Learning-based intrusion detection system for IoT, demonstrating reliable performance in detecting various attacks with an average accuracy of 93.74%, contributing to IoT network security.

Jothi and Pushpalatha [71] highlight the importance of IoT applications and the vulnerability of IoT networks to security threats. A novel intrusion detection system (IDS), WILS, based on deep learning models, outperforms existing models in accuracy, precision,

TABLE 7. Year-wise literature survey on IoT using machine learning (2020).

Authors	Key Concept	Findings
Verma et al. [43]	IoT Security against DoS Attacks	Explored ML algorithms for anomaly-based intrusion detection in IoT. Benchmarked with popular datasets, highlighted Raspberry Pi for hardware evaluation. Proposed classifier selection methodology. Emphasized the use of ensemble learning for IDS development.
Kiran et al. [44]	Machine Learning for IoT Attacks	Simulated IoT environment, conducted attacks and built classifiers for normal and attack data categorization.
Rahman et al. [45]	Semi-Distributed and Distributed IDS	Proposed semi-distributed and distributed IDS methods for resource-constrained IoT in smart cities, addressing latency issues.
Hindy et al. [46]	MQTT-Based Attacks	Evaluated ML techniques for detecting MQTT-based attacks, emphasizing flow-based features.
Amouri et al. [47]	MQTT-Based Attacks	Assessed ML techniques for MQTT-based attacks, releasing a simulated dataset. Emphasized model adequacy for MQTT-based networks.
Sadikin et al. [48]	ZigBee IDS with Hybrid Approach	Introduced hybrid IDS combining rule-based and ML-based anomaly detection for ZigBee-based IoT.
Maleh [49]	IoT Intrusion Detection in Aerospace CPS	Applied ML techniques for intrusion detection, achieving over 99% accuracy with low energy consumption.
Liu et al. [50]	Machine Learning for NSL-KDD Dataset Attacks	Studied NSL-KDD attacks, compared ML algorithms, and highlighted tree-based methods' effectiveness.
Bagaa et al. [51]	ML-Based Security Framework for IoT	Presented an ML-based framework leveraging SDN and NFV for efficient IoT attack detection in an intelligent building scenario.

TABLE 8. Year-wise literature survey on IoT using machine learning (2019).

Authors	Key Concept	Findings
Chaabouni et al. [52]	IoT Security and NIDS	Surveyed IoT security threats, evaluated defence techniques, and reviewed NIDS implementations and datasets. Covered traditional and ML-based NIDS techniques comprehensively.
Papafotikas et al. [53]	IoT Device Protection	Proposed a system with an ML-based clustering algorithm for detecting suspicious behaviour in smart IoT devices. K-Means Clustering Algorithm with Supervised Training successfully detected suspicious behavior.
Hasan et al. [54]	Machine Learning for Attack Prediction	Compared ML models for predicting attacks and anomalies. Decision Tree, Random Forest, and ANN showed promising results with 99.4% test accuracy.
Fenanir et al. [55]	Lightweight IDS for IoT	Proposed a lightweight IDS using feature selection and classification. Evaluated ML techniques and selected Decision trees for outstanding performance.
Sapre et al. [56]	Comparison of IDS Datasets	Compared KDDCup99 and NSL-KDD datasets using ML classifiers. NSL-KDD showed higher quality, achieving an average accuracy improvement of 20.18%.
Anthi et al. [57]	Supervised IDS for IoT Networks	Proposed a three-layer IDS for IoT networks using a supervised approach. Achieved high F-measure results for effective detection of IoT network attacks.
Jan et al. [58]	Lightweight DoS Attack Detection	Developed a lightweight attack detection strategy for DoS attacks in IoT networks using SVM. Showed satisfactory performance in classification accuracy and detection time.
Cassales et al. [59]	Novelty Detection for IoT IDS	Introduced three data stream novelty detection techniques for intrusion detection in IoT networks. Proposed the IDSA-IoT architecture.
Yao et al. [60]	IIoT Security and Hybrid IDS	Surveyed IDS methods and system architectures for IIoT. Proposed a hybrid IDS architecture and ML-aided detection method. Outperformed existing literature in benchmarks.
Zolanvari et al. [61]	ML and Big Data for IoT Security	Explored ML and big data analytics for IoT and IIoT security. Conducted a cyber-vulnerability assessment and presented a case study with a real-world testbed. Demonstrated the effectiveness of ML-based anomaly detection against cyber-attacks.

and recall, showcasing its effectiveness for IoT networks.

Chaganti et al. [72] proposes an LSTM-based approach with SDN support for detecting network attacks in IoT networks. The model achieves an accuracy of 0.971 in classifying attack types, offering effective defense against IoT network threats.

Bakhsh et al. [73] proposed a DL-based IDS using FFNN, LSTM, and RandNN to enhance IoT network security. Each model demonstrates unique benefits and achieves high accuracy, with the proposed technique outperforming the current state-of-the-art DL-IDS.

Keshk et al. [74] addressing the need for explainable AI in cybersecurity, this paper proposes an IDS using LSTM and a novel SPIP framework. Validated with various

datasets, the framework achieves high detection accuracy and interpretability, offering potential assistance to administrators in understanding complex attack behavior.

Tsimenidis et al. [89] present a significant paradigm shift with global interconnectivity, but it also becomes a prime target for cybercriminals. This survey explores the effectiveness of deep learning in IoT intrusion detection, categorizing various models and highlighting their contributions to securing IoT ecosystems against emerging threats.

Saba et al. [75] aims to improve lives, but security threats pose challenges. This paper advocates deep learning as an advanced method for enhancing IoT security, presenting a CNN-based approach for anomaly-based intrusion detection. The model achieves high accuracy on standard datasets,

TABLE 9. Year-wise literature survey on IoT using deep learning (2024).

Authors	Key Concept	Findings
Parisa Raoufi et. al [12]	Comprehensive review of deep learning applications in IoT, analyzing 56 articles from 2019 to April 2024.	Identified current challenges and potential breakthroughs at the intersection of deep learning and IoT, providing a roadmap for future research.
Pushpa R P et. al [67]	Development of an optimized deep learning framework for real-time intrusion and anomaly detection in IoT networks.	Achieved high accuracy (98.5%) and low latency (50 ms) in detecting anomalies, with a 40% reduction in energy consumption, enhancing security and efficiency in large-scale IoT deployments.
Mei Liu et. al [68]	Literature review and implementation of ensemble deep learning models for IoT network traffic anomaly detection.	Demonstrated over 98% accuracy in detecting anomalies using ensemble techniques, highlighting the effectiveness of deep learning in monitoring complex IoT networks.
Amrik Singh et. al [69]	Exploration of deep learning techniques for anomaly detection in IoT systems across various applications.	Discussed the integration of models like CNNs and LSTMs for real-time anomaly detection, emphasizing the need for adaptive learning techniques to enhance IoT security.
Nadia Ansar et. al [70]	Proposal of an intrusion detection system for IoT environments using a hybrid CNN-LSTM deep learning model.	Achieved 99.52% accuracy in classifying network traffic as benign or malicious, offering a robust solution for enhancing IoT network security.

TABLE 10. Year-wise literature survey on IoT using deep learning (2023).

Authors	Key Concept	Findings
Awajan A. et al. [71]	DL-Based Intrusion Detection for IoT	Proposes a DL-based intrusion detection system using a four-layer deep FC network for real-time detection of malicious traffic on IoT devices. Achieves an average accuracy of 93.74% and maintains a 93.21% detection rate, enhancing IoT network security.
Jothi B. et al. [72]	WILS-based IDS for IoT Threat Detection	Introduces a novel intrusion detection system (IDS) using WILS networks, combining LSTM advantages. Outperforms existing models in accuracy, precision, and recall, showcasing susceptibility to IoT network threats.
Chaganti R. et al. [73]	LSTM-Based Attack Detection in SDN-supported IoT	Proposes an LSTM-based approach for detecting network attacks in IoT networks supported by SDN. Achieves an accuracy of 0.971, effectively identifying and classifying various attack types in IoT environments.
Bakhsh S. A. et al. [74]	DL-Based IDS with FFNN, LSTM, and RandNN	Introduces DL-based IDS using FFNN, LSTM, and RandNN to enhance IoT network security. Achieves superior performance with high accuracy (up to 99.93%) in detecting intrusions, providing defence against cyber threats in expanding IoT networks.
Keshk M. et al. [75]	XAI Framework for Explainable IoT IDS	Proposes an XAI framework for intrusion detection in IoT networks using LSTM. Incorporates the SPIP framework for interpretability, achieving high detection accuracy, processing time, and interpretability in understanding complex attack behaviour.

TABLE 11. Year-wise literature survey on IoT using deep learning (2022).

Authors	Key Concept	Findings
Tsimenidis S. et al. [76]	Deep Learning for IoT Intrusion Detection	Survey explores deep learning in IoT intrusion detection, categorizing models and highlighting their contributions to securing IoT ecosystems.
Saba T. et al. [77]	CNN-Based Anomaly Detection for IoT Security	Proposes a CNN-based approach for anomaly-based intrusion detection in IoT, achieving high accuracy on standard datasets.
Otoum Y. et al. [78]	DL-IDS with Spider Monkey Optimization and SDPN	Introduces DL-IDS for IoT security, combining spider monkey optimization and stacked-deep polynomial networks, outperforming existing systems.
Banaamah A. M. et al. [79]	Comparative Analysis of Deep Learning Models for IoT	Focuses on cybersecurity in various applications, introduces deep learning for IoT intrusion detection, and compares CNNs, LSTMs, and GRUs.
Wahab O. A. et al. [80]	Online Deep Neural Network for Dynamic IoT Environments	Proposes an online deep neural network to address challenges in dynamic IoT environments, achieving enhanced stability in intrusion detection.
Ravi V. et al. [81]	Deep Learning in SDN-IoT for Real-Time Attack Detection	Highlights the increasing IoT attacks, introducing a deep-learning-based approach for SDN-IoT, outperforming classical models in real-time attack detection.

showcasing its potential in detecting intrusions and abnormal traffic behavior.

Otoum et al. [76] gave exponential growth of IoT applications, security has become a critical concern. The proposed DL-IDS, utilizing spider monkey optimization

and stacked-deep polynomial networks, addresses challenges in securing IoT devices. Extensive analysis demonstrates superior accuracy, precision, recall, and F-score performance compared to existing intrusion detection systems.

TABLE 12. Year-wise literature survey on IoT using deep learning (2021).

Authors	Key Concept	Findings
Ge M. et al. [82]	Customized deep learning for IoT intrusion detection	Proposed a novel intrusion detection approach for IoT using a custom deep learning technique. Utilized an IoT dataset with realistic attack traffic, achieving high classification accuracy for binary and multiclass scenarios.
Qaddoura R. et al. [83]	Deep multilayer classification for IoT intrusion detection	Proposed a deep multilayer classification approach for intrusion detection in IoT networks. Outperformed other classification techniques, achieving a higher G-mean value.
Maseer Z. K. et al. [84]	Hybrid weighted deep belief network for cyberattack detection	Introduced a novel hybrid weighted deep belief network (HW-DBN) algorithm, DeepIoT.IDS is used for efficient and reliable intrusion detection in IoT. Outperformed three recent models, achieving high detection accuracy, especially for web and bot attacks.
Abdel-Basset M. et al. [85]	Semi-supervised deep learning for IoT intrusion detection	Proposed a semi-supervised deep learning approach, SS-Deep-ID, with multiscale residual temporal convolutional module and improved traffic attention mechanism. Considering IoT traffic data's imbalanced nature demonstrated improved efficiency in intrusion detection.
Zhong M. et al. [86]	Sequential model-based intrusion detection using deep learning	Presented a sequential model-based intrusion detection system for IoT servers. Utilized Text-CNN and GRU methods, achieving a higher F1-score than traditional methods.
Selvapandian D et al. [87] & Santhosh, R.	Deep learning-based intrusion detection for multi-cloud IoT	Proposed a deep learning-based intrusion detection system for multi-cloud IoT environments. Improved detection accuracy and training efficiency compared to neural network-based models, validated through experimental evaluation with the NSL-KDD dataset.
Shareena J. et al. [88]	Deep Neural Network for IoT DDoS Botnet attack detection	Designed an intrusion detection system based on a highly extensible Deep Neural Network (DNN) for uncovering IoT DDoS Botnet attacks. Demonstrated superior accuracy and precision compared to existing systems, contributing to IoT industry security.
Fatani A et al. [89]	AI-based intrusion detection using deep learning and meta-heuristics	Proposed an efficient AI-based mechanism for intrusion detection in IoT systems. Leveraged deep learning and metaheuristics, achieving higher accuracy compared to existing approaches.
Wani A. et al. [90]	SDN-based intrusion detection using deep learning for IoT	Introduced an SDN-based intrusion detection system for IoT, utilizing a deep learning classifier. The proposed system does not burden IoT devices with security profiles, showing promising performance in simulated environments compared to other methods.

TABLE 13. Year-wise literature survey on IoT using deep learning (2020).

Authors	Key Concept	Findings
Susilo B. et al. [91]	IoT Security with Deep Learning	Proposed using machine learning, specifically deep learning, to enhance IoT security. Developed an algorithm for detecting denial-of-service (DoS) attacks, demonstrating increased accuracy in mitigating attacks on IoT networks.
Almiani M. et al. [92]	Fog Security with AI-Based IDS	Presented an AI-based intrusion detection system for Fog security against cyber-attacks. Utilized multi-layered recurrent neural networks and demonstrated stability and robustness using the NSL-KDD dataset. Measured performance with various metrics, including Mathew correlation and Cohen's Kappa coefficients.
Dawoud A. et al. [93]	Unsupervised DL for Anomaly Detection in IoT	Explored deep learning algorithms, specifically Restricted Boltzmann machines and Autoencoders, for anomaly detection in IoT. Proposed a novel framework achieving approximately 99% detection accuracy, outperforming closely related work.
Li Y. et al. [94]	Multi-CNN Fusion for Intrusion Detection	Introduced a deep learning approach for intrusion detection using a multi-convolutional neural network (multi-CNN) fusion method. Achieved superior performance on the NSL-KDD dataset compared to traditional machine learning methods.
Qiu H. et al. [95]	Adversarial Attacks on DL-Based NIDS in IoT	Investigated the vulnerability of DL-based network intrusion detection systems in IoT to adversarial attacks. Proposed an adversarial attack strategy with black-box accesses, successfully compromising a state-of-the-art NIDS. Demonstrated efficient generation of adversarial examples with a high attack success rate.
Fenanir S. et al. [96]	Lightweight IDS with Deep Auto-Encoder (DAE)	Proposed a lightweight intrusion detection system (IDS) based on a Deep Auto-Encoder (DAE) for identifying attacks in IoT due to the constrained nature of devices. Achieved promising results in accuracy, detection rate, and false alarm rate.
Liang C. et al. [97]	Hybrid Placement Strategy for IoT Intrusion Detection	Focused on designing, implementing, and testing an intrusion detection system using a hybrid strategy involving multi-agent, blockchain, and deep learning algorithms. Demonstrated the efficiency of deep learning algorithms in detecting attacks from the transport layer.
Riyaz B. et al. [98]	Feature Selection and CNN for Wireless Network IDS	Proposed a new intrusion detection system for wireless networks using a feature selection algorithm and convolutional neural network. Achieved an overall detection accuracy of 98.88%, outperforming existing systems.
Otoum Y. et al. [99]	DL Approaches for IoT Security	Explored the impact of IoT on various applications and presented detailed analyses of IoT security requirements and challenges. Reviewed state-of-art research work in IoT security domains using DL approaches, including RNN, LSTM, CNN, DBN, and AE. Identified research issues and outlined future directions for DL algorithms in IoT security.
Li B. et al. [100]	Federated Deep Learning for Industrial CPSs	Proposed DeepFed, a federated deep learning scheme for detecting cyber threats against industrial cyber-physical systems (CPSs). Utilized a convolutional neural network and a gated recurrent unit within a federated learning framework. Demonstrated high effectiveness in detecting various types of cyber threats to industrial CPSs.

Banaamah and Ahmad [77] focuses on cybersecurity in various applications, introducing deep learning for intrusion

detection in IoT. Comparative analysis of CNNs, LSTMs, and GRUs reveals the proposed method's highest accuracy among

TABLE 14. Year-wise literature survey on IoT using deep learning (2019).

Authors	Key Concept	Findings
Ge M. et al. [101]	Intrusion Detection in IoT using DNN	Proposed a novel intrusion detection scheme for IoT networks utilizing deep learning, specifically feed-forward neural networks. Demonstrated high classification accuracy for various attack types.
Thamilarasu G. et al. [102]	Real-time IoT Intrusion Detection with DL	Developed an intelligent intrusion-detection system tailored to IoT, employing a deep-learning algorithm for detecting malicious traffic. Demonstrated effective intrusion detection in both real and simulated networks.
Vinayakumar R. et al. [103]	DNNs for Flexible and Effective IDS	Explored the use of deep neural networks (DNNs) for developing a flexible and effective intrusion detection system (IDS) to detect unforeseen and unpredictable cyberattacks. Showcased DNNs' superior performance over classical classifiers.
Li D. et al. [104]	IoT Feature Extraction for Intrusion Detection	Proposed an intrusion detection algorithm based on deep migration learning for the IoT industry. Evaluated the algorithm using the KDD CUP 99 dataset, demonstrating efficient detection of anomalies.
Zhang Y et al. [105]	Adaptive IDS Using Genetic Algorithm and DBN	Presented an intrusion detection model based on an improved genetic algorithm (GA) and deep belief network (DBN), adapting its structure for different attack types. Verified effectiveness using the NSL-KDD dataset.
Pektaş A. et al. [106]	Spatial-Temporal Features in Intrusion Detection	Introduced a deep neural network model combining convolution neural network and long short-term memory for enhancing intrusion detection performance. Outperformed other publicly available methods.
Zhao L. et al. [107]	Device-Free Localization with BE-CNN	Proposed a background elimination-based convolutional neural network (BE-CNN) for device-free localization in the IoT. Showed improved accuracy and robustness over other comparison methods.
Callegari C. et al. [108]	Real-Time IDS for IoT Using Probabilistic Data Structures and DL	Proposed an intrusion detection system combining probabilistic data structures and deep learning for real-time processing of IoT network traffic. Achieved about 90% detection rate with an accuracy of 0.871.
Papamartzivanos D. et al. [109]	Self-Adaptive Misuse IDS with DL	Introduced a self-adaptive, scalable, and autonomous misuse IDS using deep-learning-based methods. Demonstrated increased attack detection rates without constant manual updates.
Zhang J. et al. [110]	Vehicle Intrusion Detection with Neural Network	Explored vehicle intrusion detection using a neural network algorithm in deep learning. Leveraged gradient descent with momentum (GDM) and gradient descent with momentum and adaptive gain (GDM/AG) for efficient anomaly detection. Achieved high accuracy in detecting unknown attacks.

existing approaches, emphasizing the potential for improved security in IoT environments.

Wahab [78] presented machine learning-based intrusion detection systems in IoT face challenges in dynamic environments. The proposed online deep neural network dynamically adjusts hidden layer sizes to counter data and concept drifts. Experimental results indicate enhanced stability compared to static models, offering a solution for effective intrusion detection in dynamic IoT environments.

Ravi et al. [79] highlight the increasing number of IoT attacks, emphasizing the significance of Software Defined Networking (SDN). The proposed deep-learning-based approach, incorporating GRU and kernel-PCA, outperforms classical models in real-time attack detection and classification in the SDN-IoT environment. The solution effectively monitors network traffic and proactively alerts about potential attacks.

Awajan [80] address the vulnerability of Internet of Things (IoT) devices to cyber attacks due to their limited resources. They propose an intrusion detection approach using a customized deep learning technique, employing a feed-forward neural network model for multiclass classification. Transfer learning is applied to build a binary classifier, demonstrating high classification accuracy for binary and multiclass scenarios.

Qaddoura et al. [81] focus on the security of IoT networks, emphasizing the impact on service availability and user privacy. The authors propose a deep multilayer classification approach for intrusion detection, incorporating two detection

stages and an oversampling technique. Experimental results show the proposed approach outperforms other classification techniques, achieving a higher G-mean value.

Abdel Basset et al. [82] address the challenge of limited labeled records for IoT intrusion detection. The authors introduce a semi-supervised deep learning approach, SS-Deep-ID. It incorporates a multiscale residual temporal convolutional module and an improved traffic attention mechanism. The proposed method enhances intrusion detection efficiency and robustness, particularly in real-time scenarios.

Maseer et al. [83] focuses on improving the accuracy and performance of Intrusion Detection Systems (IDS) for IoT servers using deep learning methods. The authors propose a sequential model-based IDS employing Text-CNN and GRU methods. Experimental results show that the deep learning methods outperform traditional approaches, contributing to the security of IoT servers. Selvapandian and Santhosh [85] address security concerns in IoT networks by proposing a deep learning-based intrusion detection system for multi-cloud IoT environments. The model aims to overcome the limitations of neural network-based intrusion detection models by improving detection accuracy and training efficiency. Experimental evaluation using the NSL-KDD dataset demonstrates improved performance.

Shareena et al. [86] focus on designing an intrusion detection system based on deep learning to uncover IoT DDoS Botnet attacks. The proposed highly extensible Deep Neural Network (DNN) demonstrates superior accuracy and

precision compared to existing systems, contributing to the security of the IoT industry.

Fatani et al. [87] address growing cyber threats in IoT systems. The authors propose an efficient AI-based mechanism for intrusion detection using deep learning and metaheuristics algorithms. The method leverages convolutional neural networks (CNNs) for feature extraction and introduces a new feature selection method, achieving higher accuracy than existing approaches.

Wani et al. [88] introduce an SDN-based intrusion detection system for IoT, leveraging a deep learning classifier to detect anomalies. This approach aims to provide security without burdening resource-constrained IoT devices with security profiles. Simulation results indicate promising performance compared to other relevant methods.

Susilo and Sari [109] discuss the increasing challenges in IoT security and propose using machine learning, specifically a deep-learning algorithm, to enhance security. The research focuses on detecting denial-of-service (DoS) attacks using a deep-learning model implemented in Python. The findings reveal that the deep-learning approach significantly improves accuracy in mitigating attacks on IoT networks.

Almiani et al. [90] address Fog and Cloud security concerns in the context of IoT, emphasizing the importance of intrusion detection systems. The proposed model, utilizing artificial intelligence and recurrent neural networks, demonstrates stability and robustness against cyber-attacks. The NSL-KDD dataset is employed for validation, and the results, measured using various metrics, confirm the model's effectiveness.

Dawoud et al. [91] focusing on the security challenges posed by IoT, the paper explores the use of deep learning algorithms, specifically Restricted Boltzmann machines and Autoencoders, for anomaly detection. The proposed unsupervised deep learning framework exhibits a high detection accuracy of approximately 99%, surpassing related work.

Li et al. [92] introduce a deep learning approach for intrusion detection using a multi-convolutional neural network (multi-CNN) fusion method. This method transforms one-dimensional feature data into grayscale graphs, and the experimental results on the NSL-KDD dataset show superior performance compared to traditional machine learning methods.

Qiu et al. [93] investigate the vulnerability of DL-based network intrusion detection systems in IoT to adversarial examples. The proposed adversarial attack utilizes black-box accesses and demonstrates the compromise of a state-of-the-art NIDS. The research introduces model extraction and a saliency map to efficiently generate adversarial examples, achieving a high attack success rate.

Fenanir et al. [94] addressing the identification of attacks in IoT due to the constrained nature of devices, the paper proposes a lightweight intrusion detection system based on a Deep Auto-Encoder (DAE). The DAE learns standard patterns and detects anomalies, providing promising results in accuracy, detection rate, and false alarm rate.

Liang et al. [95], focusing on IoT network security, research designs an intrusion detection system using a hybrid strategy involving a multi-agent system, blockchain, and deep learning algorithms. The proposed system, tested on the NSL-KDD dataset, demonstrates the efficiency of deep learning algorithms in detecting attacks from the transport layer.

Riyaz and Ganapathy [96] introduce a new intrusion detection system for wireless networks using a feature selection algorithm and convolutional neural network. The proposed model achieves an overall detection accuracy of 98.88%, showcasing improved performance compared to existing systems. Otoum and Nayak [97] explored the impact of IoT on various applications and presented the Deep Learning (DL) approaches to enhancing IoT security. A comparative analysis of DL algorithms, including RNN, LSTM, CNN, DBN, and AE, is conducted. The research identifies current issues and outlines future directions for DL algorithms in IoT security domains.

Li et al. [98], addressing the increasing attack surface of industrial cyber-physical systems (CPSs), the paper proposes DeepFed, a federated deep learning scheme. The scheme combines a convolutional neural network and a gated recurrent unit for intrusion detection in CPSs. Extensive experiments on accurate industrial CPS data demonstrate the effectiveness and superiority of DeepFed over existing schemes.

Ge et al. [99] highlights the importance of securing Internet of Things (IoT) networks against adversarial attacks due to their constrained resources. The authors propose a novel intrusion detection scheme utilizing deep learning concepts, specifically feed-forward neural networks, to classify traffic flow in IoT networks. The evaluation of the scheme using a newly published IoT dataset demonstrates high classification accuracy for various attack types.

Thamilarasu and Chawla [100] emphasizes the growing cyber-attacks on IoT networks and the need for real-time detection to mitigate their impact. The authors develop an intelligent intrusion-detection system for IoT, employing a deep-learning algorithm to detect malicious traffic. The proposed system demonstrates effective intrusion detection through real-network traces and simulations, showcasing scalability.

Vinayakumar et al. [101] addressing the challenges in developing intrusion detection systems (IDS) for evolving cyber threats, the paper explores the use of a deep neural network (DNN) to create a flexible and effective IDS. The study evaluates various machine learning algorithms on publicly available malware datasets, highlighting the superior performance of DNNs compared to classical classifiers. The proposed DNN model, called scale-hybrid-IDS-AlertNet, is suggested for real-time network traffic monitoring.

Li et al. [102] focusing on the security challenges in the Internet of Things (IoT) industry, the paper proposes an intrusion detection algorithm based on deep migration learning. This algorithm combines deep learning with intrusion detection technology, offering a solution for identifying

anomalies in IoT applications. The experimental results, utilizing the KDD CUP 99 dataset, indicate the proposed algorithm's detection time and effectiveness efficiency.

Zhang et al. [103] recognizing the evolving security concerns in the Internet of Things (IoT), the paper introduces an intrusion detection model based on an improved genetic algorithm (GA) and deep belief network (DBN). The proposed model adapts its network structure through GA iterations to enhance detection accuracy across different attack types. Evaluation using the NSL-KDD dataset confirms the model's effectiveness in improving intrusion attack recognition rates.

Pektaş and Acarman [104] presents a deep neural network model for enhancing intrusion detection performance by automatically learning spatial-temporal features from network flows. The proposed architecture, combining convolution neural network and long short-term memory, outperforms other publicly available methods to detect abnormal traffic, achieving a high accuracy of 99.09% and a low false alarm rate of 0.0227.

Zhao et al. [105] addressing device-free localization challenges in the Internet of Things (IoT) era, the paper introduces a background elimination-based convolutional neural network (BE-CNN) scheme. The proposed method improves the localization accuracy and robustness for device-free localization, showcasing superiority over comparison methods such as deep neural networks with autoencoder, K-nearest-neighbor (KNN), and support vector machines (SVM). Callegari et al. [106] focusing on the Internet of Things (IoT), this paper proposes an intrusion detection system combining probabilistic data structures and deep learning for real-time processing of network traffic. Experimental tests confirm the system's ability to achieve about 90% detection rate with an accuracy of 0.871, demonstrating its suitability for IoT security.

Papamartzivanos et al. [107] addressing the lack of agility in misuse Intrusion Detection Systems (IDSs), the paper introduces a self-adaptive, scalable, and autonomous misuse IDS using deep-learning-based methods. The proposed methodology, combining self-taught learning and MAPE-K frameworks, enhances IDS performance without constant manual updates, achieving increased attack detection rates even in dynamically changing environments.

Zhang et al. [108] explores vehicle intrusion detection using a neural network algorithm in deep learning. Utilizing gradient descent with momentum (GDM) and gradient descent with momentum and adaptive gain (GDM/AG), the proposed model demonstrates faster convergence and adaptability to detect unknown attacks in actual vehicles, achieving high accuracy in anomaly detection.

2) DEEP LEARNING ALGORITHMS IN IDS

Intrusion Detection Systems (IDS) play a vital role in identifying and responding to security threats in computer networks. Deep learning algorithms show promise in enhancing IDS capabilities by enabling the system to learn

and adapt to evolving cyber threats. The following are some commonly used deep learning algorithms in Intrusion Detection Systems [110].

It's essential to note that the effectiveness of deep learning algorithms in IDS depends on various factors, including the quality and representativeness of training data, the chosen architecture, and the specific requirements of the targeted network environment. Additionally, traditional machine learning techniques and rule-based approaches are often combined with deep learning for a more comprehensive intrusion detection system [112].

D. REINFORCEMENT LEARNING IN INTRUSION DETECTION FOR IoT SYSTEMS

The Internet of Things (IoT) has revolutionized various industries, but its widespread adoption has also introduced significant security vulnerabilities. The heterogeneous nature of IoT devices, their constrained resources, and the growing sophistication of cyberattacks make traditional intrusion detection systems (IDS) less effective. Reinforcement learning (RL), a subset of machine learning, has emerged as a promising approach for developing adaptive, autonomous, and efficient IDS. This systematic review highlights the progress in applying RL to IoT intrusion detection, summarizing key contributions and challenges.

Nguyen et al. [113] proposed an innovative approach to IoT intrusion detection by modeling it as a Markov Decision Process (MDP) to learn optimal detection policies dynamically. The study evaluated this approach in simulated IoT environments and demonstrated its effectiveness in detecting Distributed Denial-of-Service (DDoS) attacks. The MDP-based model showcased the ability to adapt to changing network traffic patterns, significantly improving detection rates compared to traditional static rule-based systems. This dynamic adaptability highlights the potential of MDPs in addressing the evolving nature of IoT security challenges.

Javaid et al. [114] proposed the use of a Deep Q-Network (DQN) for intrusion detection in IoT environments, with a focus on detecting botnet and ransomware attacks. Using the BoT-IoT dataset for evaluation, the DQN model demonstrated superior performance compared to traditional supervised learning techniques, particularly in accuracy and reduced false favorable rates. The deep reinforcement learning (DRL) approach effectively handled high-dimensional data and identified complex attack patterns, achieving an impressive accuracy of over 95% on the test set. This study highlights the potential of DQN in enhancing IoT intrusion detection systems by enabling accurate and efficient threat detection.

Zhao et al. [115] introduced a novel approach that combines federated learning with reinforcement learning (RL) to collaboratively train intrusion detection system (IDS) models across IoT devices while preserving data privacy. By avoiding sharing raw data, this method addresses privacy concerns, making it highly suitable for sensitive IoT environments.

TABLE 15. Deep learning algorithms in IDS.

Algorithms	Application	Usage in IDS
Convolutional Neural Networks (CNNs)	Widely utilized in image recognition, CNNs can also be applied to network traffic data.	CNNs effectively learn spatial hierarchies of features from network packet data, aiding in detecting patterns indicative of attacks.
Recurrent Neural Networks (RNNs)	Designed for sequential data processing, RNNs are commonly used in natural language processing tasks.	RNNs capture temporal dependencies in network traffic data, proficiently analyzing sequences of network events and detecting anomalous patterns [111].
Long Short-Term Memory (LSTM) Networks	LSTMs, a type of RNN, capture long-term dependencies in sequential data.	LSTMs are suitable for IDS, learning from the entire history of network traffic data and enabling the detection of complex attacks unfolding over time [112].
Generative Adversarial Networks (GANs)	GANs are used to generate new data samples resembling a given dataset.	GANs generate synthetic network traffic data, creating diverse datasets for training IDS models and enhancing the model's ability to generalize and detect novel attack patterns.
Autoencoders	Neural networks designed for data compression and reconstruction tasks.	Autoencoders detect anomalies by learning normal patterns in network traffic data, flagging any deviation from the learned normal behavior as a potential intrusion.
Deep Autoencoding Gaussian Mixture Model (DAGMM)	DAGMM is a hybrid model combining autoencoders and Gaussian Mixture Models.	DAGMM is effective in unsupervised anomaly detection, learning the normal behavior of the network, and identifying anomalies based on deviations from the learned model.
Attention Mechanisms	Focus on different parts of input data, allowing the model to consider relevant information selectively.	Attention mechanisms enhance neural network performance by focusing on specific features or sequences in network traffic data indicative of malicious activities [112].
Capsule Networks (CapsNets):	CapsNets overcome the limitations of traditional neural networks, especially in handling hierarchical relationships in data.	CapsNets capture complex relationships in network data, potentially improving the detection of sophisticated attacks.

The proposed federated RL model was evaluated using the IoT-23 dataset and demonstrated high accuracy in detecting botnet and command-and-control (C2) attacks. This study underscores the effectiveness of federated RL in balancing robust intrusion detection with privacy preservation, marking a significant advancement in IoT security.

1) EMERGING TRENDS

- Hybrid Models: Combining RL with supervised learning to improve performance and reduce false positives.
- Explainable RL: Enhancing the interpretability of RL models to build trust among IoT administrators.
- Lightweight RL: Developing computationally efficient RL models for resource-constrained IoT devices.
- Federated RL: Collaborative learning approaches that preserve privacy and reduce communication overhead.

Reinforcement learning has shown significant promise for IoT intrusion detection, offering adaptive and intelligent solutions to combat evolving threats. The systematic application of RL, especially in conjunction with emerging trends like federated and explainable RL, has the potential to revolutionize IoT security. However, scalability, reward design, and dataset limitations must be addressed for RL-based IDS to achieve widespread deployment. Continued research and innovation in this field are critical to securing the next generation of IoT networks, as shown in table 17.

III. COMPARATIVE ANALYSIS OF MACHINE LEARNING AND DEEP LEARNING TECHNIQUES FOR NETWORK INTRUSION DETECTION SYSTEMS

This topic presents a comprehensive analysis of various machine learning (ML) and deep learning (DL) techniques utilized in network intrusion detection systems (NIDS). We compare the effectiveness of supervised and unsupervised

learning approaches using different datasets. The study focuses on dimensionality reduction, types of attacks addressed, and the achieved accuracy of each technique.

Network security is paramount in protecting sensitive data from cyber threats. Network Intrusion Detection Systems (NIDS) play a critical role in identifying and mitigating such threats. Recently, machine learning and deep learning techniques have been increasingly applied to enhance the performance of NIDS. This paper aims to compare the effectiveness of different ML and DL architectures based on their accuracy in detecting various types of network attacks.

In the realm of machine learning techniques, SVM (Kim et al. [118]) achieved an accuracy of 91.6% on the KDD99 dataset with dimensionality reduction, effectively addressing DoS, U2R, R2L, and Probe attacks. Naive Bayes (Monika Vishwakarma et al. [119]) achieved a 98.59% accuracy on combined datasets (NSL-KDD, UNSW-NB15, CICIDS2017) with dimensionality reduction. Random Forest (Belouch M et al. [120]) attained an accuracy of 97.49% on the UNSW-NB15 dataset without employing dimensionality reduction.

In deep learning techniques, ANN (Chandak [121]) achieved a remarkable 99.93% accuracy on the KDD99 dataset with dimensionality reduction. CNN (Riyaz et al. [122]) achieved 98.88% accuracy on the KDD dataset, utilizing dimensionality reduction. RNN-LSTM (Sanchit Nayyar et al. [123]) reported a 96% accuracy on the CAIDA DDoS 2007 dataset without dimensionality reduction. RNN (C. Yin et al. [124]) achieved a high accuracy of 99.81% on the NSL-KDD dataset without dimensionality reduction.

Regarding ensemble methods, BCNN (Zhang et al. [125]) achieved a 99.12% accuracy, although the dataset and specific attacks addressed were not specified. The Adaptive ensemble machine learning (Gao et al. [126]) attained an 85.2%

TABLE 16. Comparison of various parameters of Machine learning, deep learning, and ensemble techniques.

Author	Learning Architecture	Technique	Category	Dataset	Dimensional reduction	Attacks addressed	Accuracy (%)
Huiwen Wang et al. [51]	LMRDT-SVM	Machine Learning	Supervised	NSL-KDD	Yes	No attacks	99.31
Kim et al. [50]	SVM	Machine Learning	Supervised	KDD99	Yes	DoS, U2R, R2L, Probe	91.6
Monika Vishwakarma et al. [33]	Naive Bayes Method	Machine Learning	Supervised	NSL-KDD, UNSW-NB15, and CICIDS2017	Yes	DoS, U2R, R2L, Probe	98.59
Sharmila B S et al. [35]	Naive Bayes Method	Machine Learning	Supervised	NSL-KDD	Yes	DoS, U2R, R2L, Probe	83
Belouch M et al. [37]	Random Forest	Machine Learning	Supervised	UNSWNB-15	No	-	97.49
Abdulhammed, R et al. [38]	Random Forest	Machine Learning	Supervised	AWID	Yes	802.11 MAC layer attacks	99.64
K. Samunnisa et al. et al. [23]	K-Means + RF	Machine Learning	UnSupervised	NSL-KDD	Yes	DoS, U2R, R2L, Probe	92.77
Vipin et al. [25]	K-means	Machine Learning	UnSupervised	NSL-KDD	No	DoS, U2R, R2L, Probe	82.29
Akashdeep et al. [31]	ANN	Deep Learning	Supervised	KDD99	Yes	DoS, U2R, R2L, Probe	99.93
Lin et al. [32]	CNN + Soft-max	Deep Learning	Supervised	KDD99 + NSLKDD	Yes	Smurf, Neptune, satan, ipsweep, portsweep	97.53
Riyaz et al. [17]	CNN	Deep Learning	Supervised	KDD	Yes	DoS, U2R, R2L, Probe	98.88
Sanchit Nayyar et al. [19]	RNN-LSTM	Deep Learning	Supervised	CAIDA DDos 2007	No	Botnet, Bruteforce, DoS, DDos	96
S. Sivamohan et al. [20]	RNN-BLSTM	Deep Learning	Supervised	CICIDS2017	Yes	Brute force, DoS, DDos	98.48
C. Yin et al. [21]	RNN	Deep Learning	Supervised	NSL-KDD	No	DoS, U2R, R2L, Probe	99.81
G. Liu et al. [22]	DBN	Deep Learning	Unsupervised	KDD Cup 1999	No	DoS, U2R, R2L, Probe	97.5
Farahna kian et al. [113]	AE	Deep Learning	Unsupervised	KDD Cup 1999	Yes	DoS, U2R, R2L, Probe	94.71
Zhang et al. [109]	BCNN	Ensemble Method	-	-	No	-	99.12
XIANWI et al. [110]	Adaptive Voting Algorithm	Ensemble Method	-	-	Yes	DoS, U2R, R2L, Probe	85.2
M A Jabbar et al. [91]	RF + AODE	Ensemble Method	-	-	No	-	90.51
Musbau et al. [92]	MPNN + SPO	Ensemble Method	-	-	Yes	-	95.02

accuracy with dimensionality reduction, addressing DoS, U2R, R2L, and Probe attacks.

The comparative analysis indicates that deep learning techniques generally outperform traditional machine learning methods in accuracy. Techniques like CNN, RNN, and their variants (e.g., LSTM, BLSTM) demonstrate superior performance in detecting various network attacks. Dimensionality reduction often enhances the accuracy of both ML and DL models. However, the choice of technique should be guided by the specific requirements of the NIDS, such as the type of attacks to be detected and the computational resources available.

We analyzed several research studies that applied various ML and DL techniques to NIDS. The comparison is based on the learning architecture, technique, sub-technique, dataset used, whether dimensionality reduction was applied, types of attacks addressed, and the accuracy achieved in Table 16.

The figure presents a bar chart comparing the accuracy of various learning architectures used in Intrusion Detection Systems (IDS). The vertical axis lists the studies' authors, while the horizontal axis represents the accuracy percentages achieved by their respective models. The accuracy values range from 82.29% (Vipin et al.) to 99.95% (Rahman M. A. et al.), illustrating a significant variation in performance across the models. Notably, the highest accuracy levels, exceeding 99%, are achieved by the models proposed by Rahman M. A. et al., Akashdeep et al., Lin et al., and C. Yin et al. In contrast, the models by Vipin et al., Sharmila B S et al., and XIANWI et al. exhibit relatively lower accuracy, below 90%. Most of the studies achieve an accuracy above 95%, reflecting the advancements in learning architectures that have significantly enhanced IDS performance. This comparison underscores the ongoing progress in developing highly accurate intrusion detection systems.

TABLE 17. Comparative analysis of reinforcement learning approaches for IoT intrusion detection.

Authors	Year	RL Approach	Key Contributions	Dataset/Results
Nguyen et. al [114]	2018	Markov Decision Process (MDP)	Models IoT intrusion detection as an MDP to dynamically learn detection policies. Demonstrates improved detection rates for DDoS attacks.	Simulated IoT environment; outperforms static rule-based systems in detecting anomalies.
Javaid et. al [115]	2020	Deep Q-Network (DQN)	Implements DQN for detecting botnet and ransomware attacks, outperforming traditional techniques in accuracy and false positive rates.	BoT-IoT dataset; achieves over 95% accuracy.
Zhao et. al [116]	2021	Multi-Agent RL (MARL)	Proposes a multi-agent framework for distributed IoT IDS, where agents collaborate to detect large-scale attacks such as DDoS and data exfiltration.	Distributed IoT environment; demonstrates scalability and reduced latency with improved detection rates.
Zhang et. al [117]	2022	Federated RL	Combines RL with federated learning to train IDS models collaboratively across IoT devices without sharing raw data, preserving privacy.	IoT-23 dataset; achieves high accuracy in detecting botnet and C2 attacks while ensuring privacy.
Afrah Gueriani et. al [118]	2024	Deep Reinforcement Learning (DRL)	Surveyed DRL-based IDS methods; classified into categories like DQN, healthcare, hybrid techniques	Various datasets; detailed performance metrics such as accuracy, recall, precision, FNR, FPR, and F-measure.

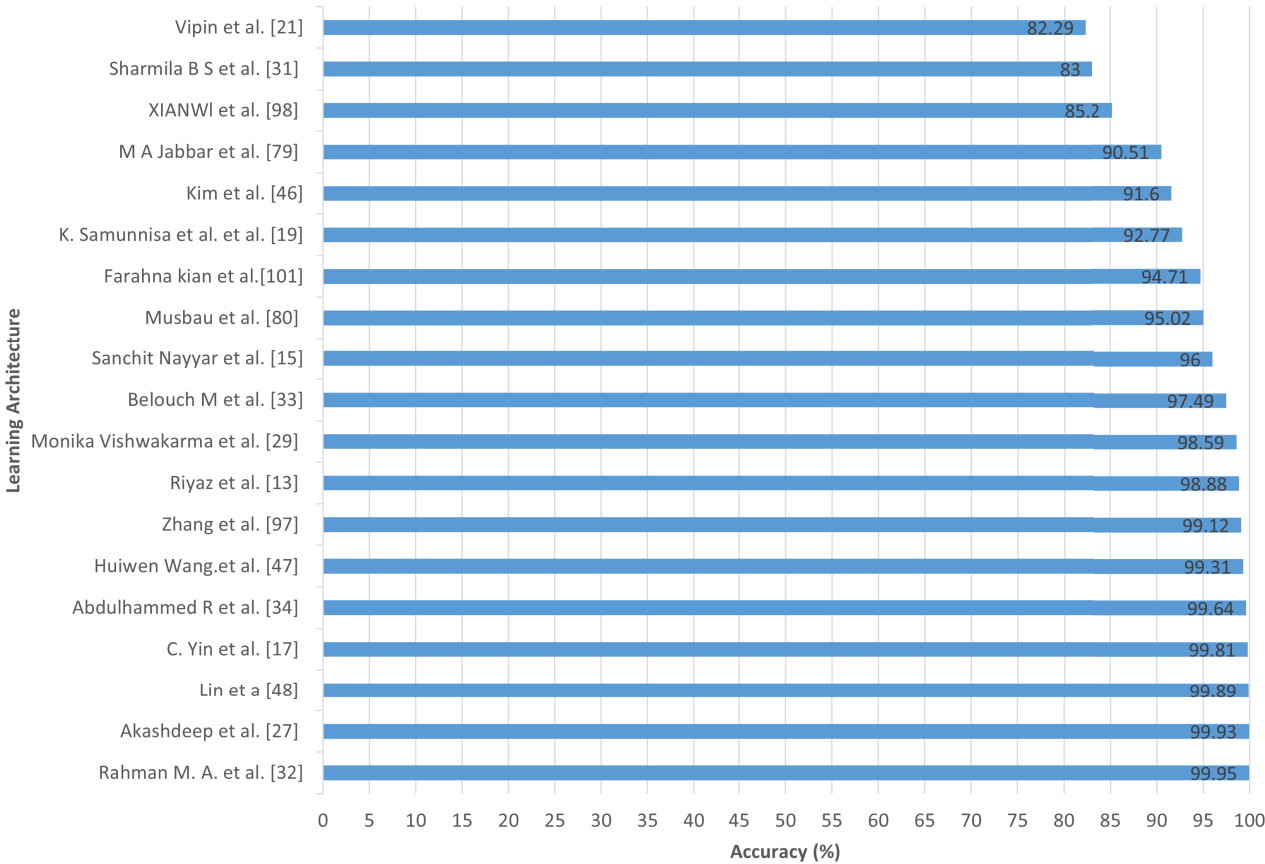


FIGURE 3. Comparison of Accuracy (%) of Different Learning Architectures for Intrusion Detection Systems illustrates the performance of various models, highlighting advancements in achieving high detection accuracy, with several models.

IV. NON-APT AND APT THREAT PATTERNS

The IoT, while offering immense technological benefits, faces significant challenges in terms of cybersecurity, primarily due to vulnerabilities in its wireless communication medium. Advanced Persistent Threats (APTs) stand out

as particularly dangerous among the various cyber threats to IoT networks. APTs are characterized by their long-term, covert, and harmful nature, making them a preferred tool for cybercriminals to compromise IoT systems. These attacks are highly sophisticated and often designed to remain

TABLE 18. Non-APT attack patterns.

Attack	CICIDS2018	AWID	UNSW-NB15	KDD99	NSL-KDD	MAWLab
DoS	✓	×	✓	✓	✓	✓
Flooding	✓	×	×	×	×	×
Fuzzers	×	×	×	×	×	✓
Heartbleed	✓	×	×	×	×	×
Brute-force	✓	✓	×	×	×	×
DDoS	✓	×	×	×	×	×
HTTP	×	×	×	×	×	✓
Multi. Points	✓	×	×	×	×	×
Alpha	×	✓	×	×	×	×
Analysis	✓	×	×	×	×	×
Generic	✓	×	×	×	×	✓

TABLE 19. APT attack patterns.

Attack	CICIDS2018	AWID	UNSW-NB15	KDD99	NSL-KDD	MAWLab
Probe	×	×	✓	✓	✓	×
Injection	×	✓	×	×	×	×
Impersonation	×	×	×	×	×	×
Reconnaissance	×	✓	×	×	×	×
R2L	×	×	×	✓	✓	×
U2R	×	×	×	✓	✓	×
Backdoors	✓	×	✓	×	×	×
Exploits	✓	×	✓	×	×	×
Shellcode	✓	×	×	×	×	×
Botnet	×	×	×	×	×	×
Web Attacks	×	×	✓	×	×	×
Worms	×	×	✓	×	×	×
Infiltration	×	×	×	×	×	×
IPv6 Tunneling	✓	×	×	×	×	×
Network Scan	✓	×	×	×	×	×
Port Scan	✓	×	✓	×	×	✓
ICMP, UDP, TCP	×	✓	×	×	×	×

undetected for extended periods, causing significant damage to networks and devices. Despite the growing reliance on Machine Learning (ML) based methods for addressing cyber threats in IoT networks, detecting APTs remains a formidable challenge. This is mainly due to the extremely low occurrence of APT attacks in network traffic, which makes it difficult for ML models to distinguish them from normal behavior and achieve reliable detection performance [118].

In contrast, non-APT threats, as in table 18, such as Denial-of-Service (DoS) attacks and other common intrusions, are more frequent and often easier to identify using traditional Intrusion Detection Systems (IDS) and ML approaches. While still harmful, these attacks lack the sophisticated and persistent characteristics of APTs, making them less challenging to detect. However, APT detection in IoT networks lacks publicly available datasets that comprehensively cover all APT attack scenarios, as shown in table 19. This scarcity of resources limits the ability of researchers to investigate and develop effective detection mechanisms for APTs.

To address these challenges, there is a growing need for a comprehensive review that bridges the gap between current network intrusion detection techniques and the detection of APTs in IoT environments. Such a review would identify the security challenges faced by IoT networks and summarize well-known attack patterns, APT-specific threat models,

and various IDS approaches, including signature-based, anomaly-based, and hybrid models. Furthermore, it would provide valuable statistical insights into the effectiveness of ML-based methods in combating network intrusions, particularly in IoT systems.

The article will also highlight several open issues and challenges in this domain. These include the lack of high-quality and diverse datasets for APT attack detection, the difficulties in applying ML approaches to rare and highly complex APTs, and the need to enhance the overall detection performance of IoT networks. Future research can focus on developing more robust, scalable, and practical solutions for securing IoT systems against APT and non-APT threats by addressing these gaps. Ultimately, this work would provide critical guidance for advancing cybersecurity in IoT networks, paving the way for more resilient and secure systems.

V. FINDINGS AND DISCUSSION

The collection of papers presented here reflects the ongoing and increasing concern about the security challenges posed by the rapid proliferation of Internet of Things (IoT) devices. The common theme across these papers is the application of deep learning techniques to develop effective intrusion detection systems (IDS) for IoT networks. Here are some

TABLE 20. Open IoT datasets available.

Dataset	Year	Open	Attack Traffic	Meta-data	Format	Realistic	Memory	Records
UNB CIC IOT 2023 [120]	2023	✓	DDoS, DoS, Reconnaissance, Web-based, Brute Force, Spoofing, and Mirai	✓	Tabular	✓	13 GB	46.7 Million
Edge-IoTSet [121]	2022	✓	DoS/DDoS attacks, Information gathering, Man-in-the-middle attacks, Injection attacks, Malware attacks	✓	Packets, Tabular	✓	124 GB	30 Million
IoT_Intrusion [122]	2021	✓	DoS, DDoS, Man-in-the-Middle (MITM), Botnet, Port Scanning, Injection Attacks, Brute Force, Malware	✓	Packets, Flow-based data, and Tabular data	✓	70 GB	15 Million
IoT-23 [123]	2020	✓	Botnet, C2 (Command and Control), Background	✓	Packets, Flow	✓	50 GB	20 Million
TON-IoT [124]	2020	✓	DDoS, DoS, OS and Service Scan, Keylogging, Data exfiltration, backdoor, DDoS, dos, injection, mitm, password, ransomware, scanning, XSS	✓	PCAP, Tabular, JSON	✓	25 GB	22 Million
MQTTTest [125]	2020	✓	SlowTe, Brute force, Malformed data, Flooding, DoS attack	✓	Packets, Tabular	✓	15 GB	1.5 Million
Bot-IoT [126]	2019	✓	DDoS, DoS, OS and Service Scan, Keylogging, Data exfiltration	✓	Packets, Tabular	✓	69 GB	72 Million
DS2OS [127]	2018	✓	DoS, Malicious Control, Malicious Operation, Spying, Wrong Setup, Scan, Data Type Probing	✓	Tabular	✓	4 GB	4 Million
N-BaIoT [128]	2018	✓	Bashlite, Mirai	✓	Tabular	✓	5.5 GB	16.5 Million
CTU-13 [129]	2014	✓	Botnet, C2, Background	✓	Packets, Malware	✓	16 GB	7 Million

general findings and discussions that can be derived from the summarized topic:

- 1) Increased Cyber Threats on IoT Networks: Many papers highlight the growing number of cyber threats and attacks targeting IoT networks. The constrained resources of IoT devices make them vulnerable to various malicious activities.
- 2) Utilization of Deep Learning for Intrusion Detection: A predominant trend is the use of deep learning algorithms, such as feed-forward neural networks, convolutional neural networks (CNNs), recurrent neural networks (RNNs), and combinations thereof, for intrusion detection in IoT networks.
- 3) Effective Detection of Various Attack Types: The proposed deep learning models demonstrate effectiveness in detecting a wide range of attack types, including denial-of-service (DoS) attacks, intrusion attempts, and anomalies in network traffic.
- 4) Performance Improvement Over Traditional Methods: Deep learning-based intrusion detection systems consistently outperform traditional machine

learning approaches. Convolutional neural networks, recurrent neural networks, and hybrid architectures enhance detection accuracy and reduce false favorable rates.

- 5) Dataset Utilization and Evaluation: Authors often employ standard datasets such as NSL-KDD and KDD CUP 99 to evaluate the performance of their proposed intrusion detection models. This allows for benchmarking and comparison of results across different studies.
- 6) Real-Time Detection and Scalability: Many papers emphasize the need for real-time intrusion detection to mitigate the impact of attacks promptly. The proposed models showcase scalability and efficiency in handling the dynamic nature of IoT networks.
- 7) Hybrid Approaches: Some papers explore hybrid approaches that combine deep learning with other techniques, such as genetic algorithms, blockchain, probabilistic data structures, and feature selection algorithms. These hybrid models aim to enhance overall system performance.

- 8) Addressing Specific IoT Security Concerns: Papers address specific challenges in IoT security, such as device-free localization, fog and cloud security, and security concerns in industrial cyber-physical systems. The proposed models demonstrate adaptability to diverse IoT applications and scenarios.
- 9) Focus on Explainability and Interpretability: Some recent papers recognize the importance of explainable artificial intelligence (AI) in the context of cybersecurity. Models incorporating LSTM and novel frameworks aim to achieve high detection accuracy and interpretability to assist administrators in understanding complex attack behaviors.
- 10) Adversarial Attacks on DL-based IDS: The vulnerability of deep learning-based intrusion detection systems to adversarial attacks is acknowledged. Some studies investigate the susceptibility of DL-based network intrusion detection systems to adversarial examples, emphasizing the need for robust models.
- 11) Future Directions and Recommendations: Several papers provide insights into future directions for research in IoT security, emphasizing the need for ongoing development in intrusion detection systems, adaptability to dynamic environments, and exploring new deep learning architectures.

In summary, the findings across these papers collectively suggest that deep learning-based intrusion detection systems have emerged as practical solutions for addressing the security challenges inherent in IoT networks. These models showcase promising results in accuracy, real-time detection, and adaptability to various IoT applications and threats. However, ongoing research is needed to address evolving security concerns and improve the robustness of these systems against adversarial attacks.

A. OPEN IoT DATASETS AVAILABLE

Open IoT datasets play a crucial role in advancing research and development in IoT security, particularly for evaluating and improving AI-based Intrusion Detection Systems (IDS). These datasets provide realistic and diverse attack scenarios, enabling researchers to design, test, and benchmark models against real-world threats. These datasets vary in format, such as packets, tabular, and flow-based data, and include metadata and realistic attack traffic for practical research in table 20. Despite their availability, challenges such as dataset imbalance, lack of standardization, and gaps in covering emerging threats remain. Open IoT datasets continue to serve as a foundation for innovation, fostering the development of robust, scalable, and adaptive security solutions for IoT environments.

VI. CONCLUSION

To summarize, this paper thoroughly examines how Artificial Intelligence (AI) techniques are applied in crafting Intrusion Detection Systems (IDS) specifically designed for Internet of Things (IoT) environments. The widespread integration

of IoT devices has brought unprecedented connectivity and notable security concerns, necessitating the implementation of effective intrusion detection mechanisms. The survey provides an in-depth analysis of contemporary AI methodologies, encompassing machine learning, deep learning, and anomaly detection algorithms, within the context of IoT-based IDS. IoT systems' unique features and vulnerabilities are underscored, emphasizing the necessity for specialized intrusion detection mechanisms. The study evaluates various AI-based approaches' performance, accuracy, and scalability, offering insights into their effectiveness in identifying and mitigating diverse cyber threats targeting IoT ecosystems. Exploring existing literature enhances our understanding of the strengths and limitations of AI-driven IDS in IoT security.

Moreover, the paper explores the challenges of deploying AI-based IDS in IoT, including resource constraints, device heterogeneity, and dynamic network topologies. By acknowledging these challenges, the paper stresses the significance of addressing them to enhance the resilience of IoT systems against evolving cybersecurity threats. The document presents potential research directions and solutions, encouraging further exploration and innovation in the field. This comprehensive survey is a valuable reference for researchers, practitioners, and policymakers seeking insights into the current landscape of AI-based intrusion detection methods in IoT security. By illuminating these approaches' strengths, limitations, and prospects, the paper contributes to ongoing efforts to fortify IoT ecosystems against cybersecurity threats' dynamic and evolving landscape.

VII. FUTURE WORK

Future directions for Intrusion Detection Systems (IDS) in IoT networks can be categorized into several key areas, providing a clear roadmap for researchers to address existing challenges and explore new opportunities. **Adversarial Resilience** is a critical area, as ML-based IDS are vulnerable to adversarial attacks that can compromise their effectiveness. Future research should focus on developing adversarial training techniques, defensive distillation, and models capable of detecting zero-day threats to improve robustness against adversarial inputs. Another pressing need is the development of **Lightweight Models**, given the resource constraints of IoT devices. Researchers should explore low-latency ML algorithms, techniques like model pruning and quantization, and federated learning to create efficient IDS that balance performance with resource usage.

Explainability in IDS is becoming increasingly crucial as ML-based solutions gain prominence. Designing interpretable ML algorithms and visualization tools and leveraging methods like SHAP or LIME can enhance transparency and trust in intrusion detection systems. Furthermore, **Real-Time Detection and Scalability** are crucial for IoT networks, as timely mitigation of attacks is essential. Future IDS should prioritize optimizing data processing speeds, ensuring scalability for large-scale IoT deployments, and

incorporating online learning techniques to dynamically adapt to new attack patterns.

Another significant challenge lies in the lack of diverse and high-quality datasets. To address this, efforts should focus on creating **Dataset Diversity and Benchmarking frameworks** that cover a wide range of attack scenarios, including rare events like Advanced Persistent Threats (APTs). Standardizing evaluation methods across datasets can also facilitate consistent benchmarking and comparison of IDS methods. Additionally, **Integration with Emerging Technologies** such as blockchain, edge computing, and 5G networks offer opportunities to enhance security, reduce latency, and improve system capabilities in handling complex IoT environments.

Future IDS can also benefit from **Multi-Modal Detection Approaches**, which combine signature-based and anomaly-based methods into hybrid models for improved accuracy. Leveraging multiple data sources, such as network traffic and device logs, alongside ensemble learning techniques can further strengthen detection mechanisms. Finally, **Privacy-Preserving IDS** is an essential focus area, especially given the increasing emphasis on data privacy. Techniques like differential privacy, homomorphic encryption, and secure multi-party computation can ensure privacy while maintaining detection performance, making IDS suitable for regulatory environments.

By addressing these future directions—spanning adversarial resilience, lightweight designs, explainability, real-time detection, dataset diversity, emerging technology integration, multi-modal detection, and privacy preservation—researchers can significantly advance the field of IoT intrusion detection systems, building robust and efficient solutions to meet the evolving demands of IoT security.

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